Chapter 1

Introduction

A pattern recognition system learns a function that maps an input vector $\mathbf{x}$ to an output $\mathbf{y}$ using a set of training data. Once trained, the system is applied to previously unseen inputs, and its ability to produce accurate outputs for data that differ from the training examples is known as *generalization*, a central goal of pattern recognition. Before being passed to the learning algorithm, raw input data are often *pre-processed* or transformed into a more useful representation, a process sometimes called *feature extraction*. Such transformations can reduce irrelevant variability, lower dimensionality, and improve computational efficiency while preserving information important for the task. The same pre-processing procedure must be applied consistently to both training and test data, and care must be taken not to discard information that is necessary for accurate prediction.

Pattern recognition problems can be broadly divided into *supervised learning* and *unsupervised learning*. In supervised learning, each training input vector $\mathbf{x}$ is paired with a corresponding target output $\mathbf{t}$, and the goal is to learn a mapping from inputs to targets. If the target belongs to one of a finite set of discrete categories, the problem is called *classification*; if the target consists of one or more continuous variables, it is called *regression*. In unsupervised learning, the training data contain only input vectors without associated target values, and the goal is instead to discover structure in the data. Typical unsupervised tasks include *clustering*, which groups similar observations, *density estimation*, which models the distribution of the data, and *dimensionality reduction*, which represents high-dimensional data in a lower-dimensional space for analysis or visualization.

Reinforcement learning is concerned with learning how an agent should act in an environment in order to maximize a cumulative reward. Unlike supervised learning, the agent is not provided with the correct output for each situation, but instead learns through interaction, trial and error, and feedback in the form of rewards. Because an action may influence not only the immediate reward but also future states and rewards, reinforcement learning often involves a *credit assignment problem*: determining which earlier actions were responsible for a later outcome. A central challenge is balancing *exploration*, in which the agent tries new actions to gain information about their consequences, and *exploitation*, in which it selects actions that are already known to produce high rewards. Effective reinforcement learning requires an appropriate trade-off between these two behaviors.

Overfitting occurs when a model becomes too closely adapted to the training data, capturing not only the underlying relationship in the data but also random fluctuations or noise. An overfitted model can therefore achieve very low training error while performing poorly on previously unseen data, indicating weak *generalization*. This typically happens when the model is unnecessarily complex relative to the amount or quality of available training data. For example, in polynomial regression, increasing the polynomial order M increases the flexibility of the fitted function. If M is made sufficiently large, the polynomial can pass exactly through all training points and drive the training error to zero. However, the resulting coefficients may become very large and the fitted curve may oscillate strongly between training points.

For a fixed model complexity, increasing the size of the training set generally reduces the risk of *overfitting*, because the model is constrained by more observations and has less freedom to fit random fluctuations in individual data points. Conversely, with limited training data, highly flexible models are more prone to overfitting. Rather than controlling complexity only by reducing the number of model parameters, a common approach is *regularization*, which modifies the training objective by adding a penalty for large parameter values. For example, L_2 regularization minimizes

$$\widetilde{E}(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}\{y(\mathbf{x}_n, \mathbf{w}) - t_n\}^2 + \frac{\lambda}{2}\|\mathbf{w}\|^2, \tag{1.1}$$

where λ controls the strength of the regularization and $\|\mathbf{w}\|^2 = \mathbf{w}^{\mathrm{T}}\mathbf{w}$ is the squared L_2 norm of the parameter vector. The penalty discourages excessively large weights and therefore limits the effective flexibility of the fitted function. This approach is known as *shrinkage methods* for linear models and is commonly referred to as *weight decay* in neural networks.

1.1 Probability Theory

Probability theory provides a mathematical framework for representing and reasoning under uncertainty. In pattern recog-

nition, uncertainty can arise from measurement noise, limited data, incomplete information, or intrinsic variability in the underlying process. By assigning probabilities to possible events or outcomes, probability theory makes it possible to quantify uncertainty and combine available information systematically. When combined with decision theory, it provides the basis for making predictions or decisions even when the available information is incomplete or ambiguous.

For discrete random variables, probabilities can be interpreted in terms of relative frequencies over repeated trials. If a random variable X can take values x_i, then $p(X = x_i)$ denotes the probability that X takes the value x_i. These probabilities satisfy

$$0 \leq p(X = x_i) \leq 1, \tag{1.2}$$

and, if the possible outcomes are mutually exclusive and exhaustive,

$$\sum_i p(X = x_i) = 1. \tag{1.3}$$

When two random variables X and Y are considered together, their joint probability $p(X, Y)$ describes the probability of particular combinations of their values. For discrete variables, $p(X = x_i, Y = y_j)$ denotes the probability that $X = x_i$ and $Y = y_j$ occur simultaneously. The probability distribution of one variable alone can be obtained by summing the joint distribution over all possible values of the other variable. This is called *marginalization*, and gives the *sum rule*

$$p(X) = \sum_Y p(X, Y). \tag{1.4}$$

The resulting $p(X)$ is called the *marginal probability distribution* of X.

A *conditional probability* describes the probability of one variable given that the value of another variable is known. The conditional probability of Y given X is written $p(Y \mid X)$. The joint and conditional distributions are related through the *product rule*

$$p(X, Y) = p(Y \mid X)p(X). \tag{1.5}$$

Equivalently, by interchanging the roles of X and Y,

$$p(X, Y) = p(X \mid Y)p(Y). \tag{1.6}$$

The sum rule and product rule form the basic algebra of probability and allow more complicated probability distributions to be constructed and manipulated.

Combining the two forms of the product rule gives *Bayes' theorem*

$$p(Y \mid X) = \frac{p(X \mid Y)p(Y)}{p(X)}. \tag{1.7}$$

Using the sum rule, the denominator can be written as

$$p(X) = \sum_Y p(X \mid Y)p(Y). \tag{1.8}$$

Thus, Bayes' theorem can also be expressed as

$$p(Y \mid X) = \frac{p(X \mid Y)p(Y)}{\sum_Y p(X \mid Y)p(Y)}. \tag{1.9}$$

The denominator acts as a normalization factor ensuring that the posterior probabilities over all possible values of Y sum to one.

Bayes' theorem provides a general mechanism for updating uncertainty when new information becomes available. The distribution $p(Y)$ represents knowledge about Y before observing X and is called the *prior probability*. After observing X, the updated distribution $p(Y \mid X)$ is called the *posterior probability*. The quantity $p(X \mid Y)$ describes how likely the observed data X are under a particular value of Y and is commonly called the *likelihood*. Bayes' theorem therefore expresses the general relationship

$$\text{posterior} \propto \text{likelihood} \times \text{prior}. \tag{1.10}$$

The posterior combines prior information with evidence supplied by the observed data.

Two random variables X and Y are said to be *independent* if knowledge of one provides no information about the other. In this case, their joint probability factorizes as

$$p(X, Y) = p(X)p(Y). \tag{1.11}$$

Using the product rule in Eqs. (1.5) and (1.6), independence also implies

$$p(Y \mid X) = p(Y), \tag{1.12}$$

and similarly,

$$p(X \mid Y) = p(X). \tag{1.13}$$

Thus, for independent variables, conditioning on one variable does not alter the probability distribution of the other.

1.2 Probability densities

For a continuous random variable x, probabilities are described by a *probability density function* $p(x)$ rather than by assigning probabilities directly to individual values. For an infinitesimal interval $(x, x + \delta x)$, the probability that the variable lies within that interval is approximately $p(x)\delta x$. More generally, the probability that x lies in an interval (a, b) is

$$p(x \in (a, b)) = \int_a^b p(x)\, dx. \tag{1.14}$$

A probability density must satisfy two fundamental conditions. First, it must be nonnegative everywhere,

$$p(x) \geq 0, \tag{1.15}$$

and second, it must be normalized so that the total probability over the entire domain is one,

$$\int_{-\infty}^{\infty} p(x)\,dx = 1. \tag{1.16}$$

Unlike probabilities themselves, the value of a probability density at a particular point is not a probability. For a continuous variable, the probability of observing exactly one particular value is zero; meaningful probabilities are obtained by integrating the density over intervals.

Under a change of variables, probability must be conserved. Suppose x and y are related by a differentiable, one-to-one transformation $x = g(y)$ (g may be linear or nonlinear). For a small positive increment $\delta y > 0$, the interval $(y, y + \delta y)$ corresponds to an interval $(x, x + \delta x)$, where

$$\delta x = g(y + \delta y) - g(y). \tag{1.17}$$

The length of the corresponding interval in x is therefore $|\delta x|$. Since probability mass equals density times interval length, conservation of probability gives

$$p_y(y)\,\delta y \approx p_x(x)\,|\delta x|. \tag{1.18}$$

Dividing by δy gives

$$p_y(y) \approx p_x(x)\left|\frac{\delta x}{\delta y}\right|. \tag{1.19}$$

Taking the limit as $\delta y \to 0$,

$$p_y(y) = \lim_{\delta y \to 0} p_x(x)\left|\frac{\delta x}{\delta y}\right| = p_x(x)\left|\frac{dx}{dy}\right|. \tag{1.20}$$

Since $x = g(y)$, this can be written as

$$p_y(y) = p_x(g(y))\left|g'(y)\right|. \tag{1.21}$$

The factor $|dx/dy|$ is the *Jacobian factor*. It accounts for the stretching or compression of intervals under the transformation. If the transformation stretches an interval, the density must decrease; if it compresses an interval, the density must increase, so that the total probability mass remains unchanged. Here, the role of the absolute value is evident, since a decreasing transformation gives $dx/dy < 0$, whereas a probability density must remain nonnegative.

This also explains why a probability density is not itself a probability. The invariant quantity is the probability mass $p_x(x)\,dx$, not the numerical value of $p_x(x)$. Consequently, the shape and height of a density depend on the variable used to represent it. In particular, the location of the maximum of a density, or its *mode*, is generally not preserved under a nonlinear transformation. If $\hat{x}$ is a mode of $p_x(x)$, so that

$$\left.\frac{dp_x(x)}{dx}\right|_{x=\hat{x}} = 0, \tag{1.22}$$

then the transformed density is $p_y(y) = p_x(g(y))|g'(y)|$. Its stationary points satisfy

$$\frac{d}{dy}\left[p_x(g(y))|g'(y)|\right] = 0, \tag{1.23}$$

which contains the Jacobian. For a nonlinear transformation g, the transformed mode is not the inverse transformation of the original mode, i.e., $\hat{y} \neq g^{-1}(\hat{x})$. For a linear transformation, however, $g'(y)$ is constant, so the Jacobian only rescales the density and does not change the corresponding location of its maximum.

So far, we have assumed that the transformation g is one-to-one over the domain. If a transformation is not one-to-one, its domain must be partitioned into monotonic branches, and the probability contributions from all branches that map to the same transformed value must be added up. More generally, if $y = h(x)$ and the equation $y = h(x)$ has inverse solutions $x_i(y)$, then

$$p_y(y) = \sum_i p_x\left(x_i(y)\right)\left|\frac{dx_i}{dy}\right|. \tag{1.24}$$

For example, if $y = x^2$, then a positive value of y can arise from both $x = \sqrt{y}$ and $x = -\sqrt{y}$, so both branches contribute to the density of y. This branch-summing rule is the general form of probability conservation for non-one-to-one transformations.

The *cumulative distribution function* (CDF) gives the probability that a continuous random variable is less than or equal to a specified value. It is defined by

$$P(z) = \int_{-\infty}^{z} p(x)\,dx. \tag{1.25}$$

The probability density can be recovered from the cumulative distribution by differentiation,

$$p(x) = \frac{dP(x)}{dx}. \tag{1.26}$$

If we have several continuous variables $x_1, \ldots, x_D$, denoted collectively by the vector $\mathbf{x}$, then we can define a joint probability density $p(\mathbf{x}) = p(x_1, \ldots, x_D)$ such that the probability of $\mathbf{x}$ falling in an infinitesimal volume $\delta\mathbf{x}$ containing the point $\mathbf{x}$ is given by $p(\mathbf{x})\delta x$. This multivariate probability density must satisfy

$$p(\mathbf{x}) \geq 0 \tag{1.27}$$

and

$$\int p(\mathbf{x})\,d\mathbf{x} = 1, \tag{1.28}$$

where the integral is taken over the entire domain of $\mathbf{x}$.

The sum and product rules of probability extend naturally from discrete variables to continuous variables. For two continuous random variables x and y, the marginal density of x is obtained by integrating the joint density over all possible values of y,

$$p(x) = \int p(x, y)\, dy. \tag{1.29}$$

Thus, marginalization corresponds to summation for discrete variables and integration for continuous variables, while the product rule retains the same conceptual form. Bayes' theorem likewise applies to probability densities,

$$p(y \mid x) = \frac{p(x \mid y)p(y)}{p(x)}, \tag{1.30}$$

with the denominator obtained by integrating over the possible values of y,

$$p(x) = \int p(x \mid y)p(y)\, dy. \tag{1.31}$$

Hence, the same basic probability rules apply to discrete distributions, continuous densities, and combinations of discrete and continuous random variables; the main difference is that summations over discrete states are replaced by integrations over continuous domains.

1.3 Expectations and covariances

The *expectation* of a function $f(x)$ is its probability-weighted average under the distribution of x. For a discrete random variable,

$$\mathbb{E}[f] = \sum_x p(x)f(x), \tag{1.32}$$

whereas for a continuous random variable,

$$\mathbb{E}[f] = \int p(x)f(x)\, dx. \tag{1.33}$$

Thus, values of $f(x)$ that occur in regions of higher probability contribute more strongly to the average.

If N independent samples $x_1, \ldots, x_N$ are drawn from the distribution of x, the expectation can be approximated by the empirical average

$$\mathbb{E}[f] \approx \frac{1}{N} \sum_{n=1}^{N} f(x_n). \tag{1.34}$$

As $N \to \infty$, this approximation converges to the true expectation under suitable conditions.

When a function depends on several random variables, a subscript can be used to indicate which variable is being averaged over. For example, $\mathbb{E}_x[f(x, y)]$ denotes expectation with respect to x, treating y as fixed. The resulting quantity is therefore a function of y.

A *conditional expectation* is an expectation taken with respect to a conditional distribution. For a discrete variable,

$$\mathbb{E}_x[f \mid y] = \sum_x p(x \mid y)f(x). \tag{1.35}$$

For a continuous variable,

$$\mathbb{E}_x[f \mid y] = \int p(x \mid y)f(x)\, dx. \tag{1.36}$$

This quantity represents the average value of $f(x)$ under the conditional distribution of x given the observed value of y.

The *variance* measures the average squared deviation of a random quantity from its mean. For a function $f(x)$,

$$\operatorname{var}[f] = \mathbb{E}\left[(f(x) - \mathbb{E}[f(x)])^2\right]. \tag{1.37}$$

Expanding the square gives the equivalent form

$$\operatorname{var}[f] = \mathbb{E}[f(x)^2] - \mathbb{E}[f(x)]^2. \tag{1.38}$$

In particular, for the random variable x itself,

$$\operatorname{var}[x] = \mathbb{E}[x^2] - \mathbb{E}[x]^2. \tag{1.39}$$

Variance is always nonnegative and quantifies the spread of a random variable around its mean.

For two random variables x and y and functions $f(x)$ and $g(y)$, the *covariance* measures the extent to which $f(x)$ and $g(y)$ vary together. It is defined as

$$\operatorname{cov}[f(x), g(y)] = \mathbb{E}_{x,y}\left[(f(x) - \mathbb{E}[f(x)])\,(g(y) - \mathbb{E}[g(y)])\right]. \tag{1.40}$$

Expanding the product gives

$$\operatorname{cov}[f(x), g(y)] = \mathbb{E}_{x,y}[f(x)g(y)] - \mathbb{E}[f(x)]\mathbb{E}[g(y)]. \tag{1.41}$$

A positive covariance indicates that $f(x)$ and $g(y)$ tend to deviate from their respective means in the same direction, whereas a negative covariance indicates that they tend to deviate in opposite directions.

As a particular case, taking $f(x) = x$ and $g(y) = y$ gives

$$\operatorname{cov}[x, y] = \mathbb{E}_{x,y}\left[(x - \mathbb{E}[x])\,(y - \mathbb{E}[y])\right], \tag{1.42}$$

or equivalently,

$$\operatorname{cov}[x, y] = \mathbb{E}_{x,y}[xy] - \mathbb{E}[x]\,\mathbb{E}[y]. \tag{1.43}$$

If x and y are independent, then $f(x)$ and $g(y)$ are also independent, and therefore

$$\mathbb{E}_{x,y}[f(x)g(y)] = \mathbb{E}[f(x)]\mathbb{E}[g(y)], \tag{1.44}$$

which implies

$$\operatorname{cov}[f(x), g(y)] = 0. \tag{1.45}$$

In particular, $\operatorname{cov}[x, y] = 0$ when x and y are independent. However, zero covariance does not generally imply independence.

For random vectors $\mathbf{x}$ and $\mathbf{y}$, covariance generalizes to a matrix,

$$\operatorname{cov}[\mathbf{x}, \mathbf{y}] = \mathbb{E}_{\mathbf{x},\mathbf{y}}\left[(\mathbf{x} - \mathbb{E}[\mathbf{x}])\,(\mathbf{y} - \mathbb{E}[\mathbf{y}])^{\mathrm{T}}\right]. \tag{1.46}$$

Equivalently,

$$\mathrm{cov}[\mathbf{x}, \mathbf{y}] = \mathbb{E}[\mathbf{x}\mathbf{y}^{\mathrm{T}}] - \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{y}]^{\mathrm{T}}. \tag{1.47}$$

If we consider the covariance of the components of a vector $\mathbf{x}$ with each other, then we use a slightly simpler notation $\mathrm{cov}[\mathbf{x}, \mathbf{x}] \equiv \mathrm{cov}[\mathbf{x}], \mathbf{x}]$.

1.4 Bayesian probabilities

The *frequentist* interpretation of probability associates probability with the long-run frequency of outcomes in repeated experiments. This view is natural when the same random experiment can be repeated many times. The *Bayesian* interpretation is more general and uses probability to represent uncertainty about quantities that may not be associated with repeatable events. In this view, probability expresses a degree of belief that can be updated when new evidence becomes available.

This distinction is particularly useful in pattern recognition, where uncertainty may concern not only the observed data but also unknown model parameters. Suppose a model is described by a parameter vector $\mathbf{w}$ and the observed data are denoted by $\mathcal{D}$. Before observing the data, uncertainty about $\mathbf{w}$ is represented by the *prior distribution* $p(\mathbf{w})$, which may also be viewed as the generating distribution from which the model parameters are initialized. The likelihood $p(\mathcal{D} \mid \mathbf{w})$ assigns a score to each parameter value $\mathbf{w}$ according to how plausible the observed data $\mathcal{D}$ are under that choice of $\mathbf{w}$. After observing $\mathcal{D}$, the prior and likelihood are combined through Bayes' theorem to obtain the *posterior distribution* $p(\mathbf{w} \mid \mathcal{D})$, which represents the updated uncertainty about the model parameters,

$$p(\mathbf{w} \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \mathbf{w})p(\mathbf{w})}{p(\mathcal{D})}. \tag{1.48}$$

For example, consider the linear model

$$y = wx + \epsilon, \tag{1.49}$$

where

$$\epsilon \sim \mathcal{N}(0, \sigma^2). \tag{1.50}$$

For a given input x_i and parameter w, the model implies

$$p(y_i \mid x_i, w) = \mathcal{N}(y_i \mid wx_i, \sigma^2). \tag{1.51}$$

For fixed w, this describes the distribution of possible observations y_i. Once an observation y_i is obtained, the same expression can instead be viewed as a function of w, indicating how well each value of w fits the observed data. Assuming data points are independent, then for the data set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, the likelihood is

$$p(\mathcal{D} \mid w) = \prod_{i=1}^{N} p(y_i \mid x_i, w). \tag{1.52}$$

Each conditional density has the form

$$p(y_i \mid x_i, w) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - wx_i)^2}{2\sigma^2}\right]. \tag{1.53}$$

Substituting this Gaussian density into the likelihood gives

$$\begin{aligned} p(\mathcal{D} \mid w) &= \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y_i - wx_i)^2}{2\sigma^2}\right] \\ &= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^N \exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{N}(y_i - wx_i)^2\right]. \end{aligned} \tag{1.54}$$

Hence, the likelihood assigned to w depends directly on the total squared residual

$$\sum_{i=1}^{N}(y_i - wx_i)^2. \tag{1.55}$$

A value of w that produces predictions wx_i closer to the observations y_i has a smaller total squared residual and therefore a larger likelihood. Conversely, a value of w that produces larger residuals has a smaller likelihood. Thus, $p(\mathcal{D} \mid w)$ provides a quantitative score for how compatible each value of w is with the observed data.

Because the likelihood is not a probability distribution over $\mathbf{w}$. Therefore, in general,

$$\int p(\mathcal{D} \mid \mathbf{w})\, d\mathbf{w} \neq 1. \tag{1.56}$$

Instead, for fixed $\mathbf{w}$, $p(\mathcal{D} \mid \mathbf{w})$ is a probability distribution over the possible data $\mathcal{D}$, and therefore

$$\int p(\mathcal{D} \mid \mathbf{w})\, d\mathcal{D} = 1. \tag{1.57}$$

Given this definition of likelihood, we can state Bayes' theorem in words

$$\text{posterior} \propto \text{likelihood} \times \text{prior}. \tag{1.58}$$

The denominator $p(\mathcal{D})$ in Eq. 1.48 is called the *evidence* or *marginal likelihood*. It is obtained by marginalizing the joint distribution $p(\mathcal{D}, \mathbf{w})$ over the parameter space. Using the product rule,

$$p(\mathcal{D}, \mathbf{w}) = p(\mathcal{D} \mid \mathbf{w})p(\mathbf{w}), \tag{1.59}$$

and therefore

$$p(\mathcal{D}) = \int p(\mathcal{D}, \mathbf{w})\, d\mathbf{w} = \int p(\mathcal{D} \mid \mathbf{w})p(\mathbf{w})\, d\mathbf{w}. \tag{1.60}$$

Substituting this expression into Bayes' theorem gives

$$p(\mathbf{w} \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \mathbf{w})p(\mathbf{w})}{\int p(\mathcal{D} \mid \mathbf{w})p(\mathbf{w})\, d\mathbf{w}}. \tag{1.61}$$

The evidence therefore acts as the normalization constant for the product of the likelihood and prior, ensuring that the posterior distribution $p(\mathbf{w} \mid \mathcal{D})$ integrates to one.

The Bayesian and frequentist viewpoints therefore treat model parameters differently. In the frequentist approach, $\mathbf{w}$ is regarded as a fixed but unknown quantity, and uncertainty is characterized through the behavior of an estimator over multiple hypothetical data sets (one true $\mathbf{w}$, many possible $\mathcal{D}$). In the Bayesian approach, the observed data set is fixed, and uncertainty about $\mathbf{w}$ is represented directly by the posterior distribution $p(\mathbf{w} \mid \mathcal{D})$ (one true $\mathcal{D}$, many possible $\mathbf{w}$).

A widely used frequentist estimator is *maximum likelihood*, in which $\mathbf{w}$ is set to the value that maximizes the likelihood function $p(\mathcal{D} \mid \mathbf{w})$. This corresponds to choosing the value of $\mathbf{w}$ for which the probability of the observed data set is maximized. Hence,

$$\mathbf{w}_{\mathrm{ML}} = \arg\max_{\mathbf{w}} p(\mathcal{D} \mid \mathbf{w}). \tag{1.62}$$

Because the logarithm is monotonically increasing, this is equivalent to maximizing the log-likelihood,

$$\mathbf{w}_{\mathrm{ML}} = \arg\max_{\mathbf{w}} \log p(\mathcal{D} \mid \mathbf{w}), \tag{1.63}$$

or minimizing the negative log-likelihood,

$$\mathbf{w}_{\mathrm{ML}} = \arg\min_{\mathbf{w}} \{-\log p(\mathcal{D} \mid \mathbf{w})\}. \tag{1.64}$$

In general, maximizing likelihood is equivalent to minimizing the discrepancy between predictions and observations. For a Gaussian model, Eq. 1.54 shows that it corresponds to minimizing the sum of squared residuals. This explains why the negative log-likelihood commonly appears as a loss function in machine learning.

In frequentist inference, one approach to estimating uncertainty in $\mathbf{w}$ is to examine how the estimator varies across possible data sets. One practical method is the *bootstrap*. Given a data set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$, a new data set is created by sampling N points from $\mathcal{D}$ with replacement. Repeating this procedure L times produces L bootstrap data sets, and the variability of the corresponding parameter estimates can be used to assess statistical uncertainty.

One criticism of the Bayesian approach is that the prior distribution $p(\mathbf{w})$ may sometimes be chosen for mathematical or computational convenience rather than because it closely reflects genuine prior knowledge. Since the posterior depends directly on the prior through Eq. 1.61, different reasonable choices of $p(\mathbf{w})$ can lead to different posterior distributions, especially when the available data are limited. This dependence introduces a degree of subjectivity into Bayesian inference. To reduce the influence of the prior, *noninformative priors* are sometimes used. These priors are designed to place relatively little preference on particular parameter values before the data are observed, so that the likelihood has a stronger influence on the resulting posterior.

One criticism of the Bayesian approach is that the prior distribution $p(\mathbf{w})$ may be chosen for mathematical convenience rather than because it closely reflects genuine prior knowledge. Since the posterior depends on the prior by Eq. 1.61, different choices of $p(\mathbf{w})$ can lead to different posterior distributions, which introduces a degree of subjectivity, especially when the available data are limited. *Noninformative priors* are sometimes used to reduce this dependence by assigning relatively little preference to particular parameter values before the data is observed. Their goal is to let the likelihood dominate the posterior, with minimal influence from the prior.

That being said, constructing a truly noninformative prior is difficult because even a seemingly weak prior can still encode assumptions. For example, a Gaussian prior $p(w) = \mathcal{N}(0, \sigma^2)$ with a very large σ^2 may appear nearly flat over a wide range, but it still favors values near zero and assigns decreasing probability to increasingly large values of $|w|$. Similarly, a uniform prior is only uniform over a chosen range, so changing variables or changing the bounds can alter what is considered "uninformative". These choices become especially important in model comparison because Bayesian model evidence integrates the likelihood over the prior distribution, i.e., $p(\mathcal{D}) = \int p(\mathcal{D} \mid \mathbf{w}) p(\mathbf{w})\, d\mathbf{w}$. As a result, the width and shape of the prior can substantially affect the marginal likelihood, so two models with similar fits to the data may receive different evidence values simply because their prior distributions allocate probability mass differently across parameter space. Moreover, a poorly chosen prior can lead to misleading posterior conclusions, especially when the available data are limited and the likelihood is not strong enough to overwhelm the prior. In such cases, the posterior may become highly concentrated around parameter values favored by the prior, giving the appearance of strong certainty even when that certainty is largely driven by an inappropriate prior assumption. Frequentist evaluation methods provide a useful external check because they assess predictive performance without relying on a prior specification. Techniques such as cross-validation are therefore valuable for detecting poor generalization and for comparing models based on their performance on held-out data.

The main practical difficulty of Bayesian inference is computational. Rather than selecting a single parameter value $\mathbf{w}$, Bayesian inference maintains a posterior distribution $p(\mathbf{w} \mid \mathcal{D})$ over plausible parameter values. Consequently, prediction requires accounting for uncertainty across the entire parameter space. For a new input $\mathbf{x}_*$, the posterior predictive distribution is obtained by marginalizing over $\mathbf{w}$. Starting from the law of total probability and applying the product rule,

$$\begin{aligned} p(\mathbf{y}_* \mid \mathbf{x}_*, \mathcal{D}) &= \int p(\mathbf{y}_*, \mathbf{w} \mid \mathbf{x}_*, \mathcal{D})\, d\mathbf{w} \\ &= \int p(\mathbf{y}_* \mid \mathbf{x}_*, \mathbf{w}, \mathcal{D}) p(\mathbf{w} \mid \mathbf{x}_*, \mathcal{D})\, d\mathbf{w}. \end{aligned} \tag{1.65}$$

Given $\mathbf{w}$ and the new input $\mathbf{x}_*$, the prediction $\mathbf{y}_*$ is assumed to be conditionally independent of the training data $\mathcal{D}$, so

$$p(\mathbf{y}_* \mid \mathbf{x}_*, \mathbf{w}, \mathcal{D}) = p(\mathbf{y}_* \mid \mathbf{x}_*, \mathbf{w}). \tag{1.66}$$

In addition, treating $\mathbf{x}_*$ as a fixed new input means that it provides no additional information about $\mathbf{w}$ beyond the training data, so

$$p(\mathbf{w} \mid \mathbf{x}_*, \mathcal{D}) = p(\mathbf{w} \mid \mathcal{D}). \tag{1.67}$$

Substituting Eqs. 1.66–1.67 into Eq. 1.65, we obtain

$$p(\mathbf{y}_* \mid \mathbf{x}_*, \mathcal{D}) = \int p(\mathbf{y}_* \mid \mathbf{x}_*, \mathbf{w}) p(\mathbf{w} \mid \mathcal{D})\, d\mathbf{w}. \tag{1.68}$$

Thus, Bayesian prediction does not rely on a single parameter estimate. Instead, it averages (or integrates) the predictive distribution associated with every possible $\mathbf{w}$, weighted by its posterior probability. When $\mathbf{w}$ is high-dimensional, this averaging requires integration over a correspondingly high-dimensional parameter space, which can make exact Bayesian prediction computationally expensive or intractable.

The development of sampling methods, such as Markov chain Monte Carlo (discussed in Chapter 11) along with dramatic improvements in the speed and memory capacity of computers, opened the door to the practical use of Bayesian techniques in an impressive range of problem domains. Monte Carlo methods are very flexible and can be applied to a wide range of models. However, they are computationally intensive and have mainly been used for small-scale problems. More recently, highly efficient deterministic approximation schemes such as variational Bayes and expectation propagation (discussed in Chapter 10) have been developed. These offer a complementary alternative to sampling methods and have allowed Bayesian techniques to be used in large-scale applications

1.5 The Gaussian distribution

Various proability distributions and their key properties will be discuss extensively in Chapter 2. This section is to introduce here the most important probability distribution for continuous variables, called the *normal* or *Gaussian* distribution.

For a scalar random variable x, the Gaussian density is defined by

$$\mathcal{N}(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]. \tag{1.69}$$

which is governed by two parameters: μ, called the *mean*, and σ^2, called the variance. The square root of the variance, given by σ, is called the *standard deviation*, and the reciprocal of the variance, written as $\beta = \frac{1}{\sigma^2}$, is called the *precision*.

The Gaussian distribution satisfies the usual requirements for a continuous probability density,

$$\mathcal{N}(x; \mu, \sigma^2) > 0 \tag{1.70}$$

and

$$\int_{-\infty}^{\infty} \mathcal{N}(x; \mu, \sigma^2)\, dx = 1. \tag{1.71}$$

We can readily find expectations (its first moment) of functions of x under the Gaussian distribution. In particular, the average value of x is given by

$$\mathbb{E}[x] = \int_{-\infty}^{\infty} x\, \mathcal{N}(x; \mu, \sigma^2)\, dx = \mu, \tag{1.72}$$

which is why μ is called the mean. Its second moment is

$$\mathbb{E}[x^2] = \int_{-\infty}^{\infty} x^2\, \mathcal{N}(x; \mu, \sigma^2)\, dx = \mu^2 + \sigma^2. \tag{1.73}$$

Using the change of variable $z = x - \mu$, so that $x = z + \mu$ and $dx = dz$, we obtain

$$\begin{aligned} \mathbb{E}[x] &= \int_{-\infty}^{\infty} (z+\mu)\, \mathcal{N}(z; 0, \sigma^2)\, dz \\ &= \int_{-\infty}^{\infty} z\, \mathcal{N}(z; 0, \sigma^2)\, dz + \mu \int_{-\infty}^{\infty} \mathcal{N}(z; 0, \sigma^2)\, dz. \end{aligned} \tag{1.74}$$

Because $\mathcal{N}(z; 0, \sigma^2)$ is symmetric about zero, $z\,\mathcal{N}(z; 0, \sigma^2)$ is an odd function, and therefore

$$\int_{-\infty}^{\infty} z\, \mathcal{N}(z; 0, \sigma^2)\, dz = 0. \tag{1.75}$$

Also, any density function should integrate to one, therefore

$$\int_{-\infty}^{\infty} \mathcal{N}(z; 0, \sigma^2)\, dz = 1. \tag{1.76}$$

It follows that

$$\mathbb{E}[x] = 0 + \mu = \mu, \tag{1.77}$$

which is why μ is called the mean. Similarly, its second moment is

$$\mathbb{E}[x^2] = \int_{-\infty}^{\infty} x^2\, \mathcal{N}(x \mid \mu, \sigma^2)\, dx. \tag{1.78}$$

Keep using the change of variable $z = x - \mu$, we obtain

$$\begin{aligned} \mathbb{E}[x^2] &= \int_{-\infty}^{\infty} (z+\mu)^2\, \mathcal{N}(z; 0, \sigma^2)\, dz \\ &= \int_{-\infty}^{\infty} \left(z^2 + 2\mu z + \mu^2\right) \mathcal{N}(z; 0, \sigma^2)\, dz \\ &= \int_{-\infty}^{\infty} z^2\, \mathcal{N}(z; 0, \sigma^2)\, dz + 2\mu \int_{-\infty}^{\infty} z\, \mathcal{N}(z; 0, \sigma^2)\, dz \\ &\quad + \mu^2 \int_{-\infty}^{\infty} \mathcal{N}(z; 0, \sigma^2)\, dz. \end{aligned} \tag{1.79}$$

As shown above,

$$\int_{-\infty}^{\infty} z\, \mathcal{N}(z; 0, \sigma^2)\, dz = 0 \tag{1.80}$$

and

$$\int_{-\infty}^{\infty} \mathcal{N}(z; 0, \sigma^2)\, dz = 1. \tag{1.81}$$

For the remaining term, using

$$\mathcal{N}(z;0,\sigma^2) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{z^2}{2\sigma^2}\right), \tag{1.82}$$

we have

$$\int_{-\infty}^{\infty} z^2\,\mathcal{N}(z;0,\sigma^2)\,dz = \frac{1}{\sqrt{2\pi}\sigma}\int_{-\infty}^{\infty} z^2\exp\left(-\frac{z^2}{2\sigma^2}\right)\,dz. \tag{1.83}$$

Since

$$\frac{d}{dz}\exp\left(-\frac{z^2}{2\sigma^2}\right) = -\frac{z}{\sigma^2}\exp\left(-\frac{z^2}{2\sigma^2}\right), \tag{1.84}$$

we can write

$$z^2\exp\left(-\frac{z^2}{2\sigma^2}\right) = -\sigma^2 z\frac{d}{dz}\exp\left(-\frac{z^2}{2\sigma^2}\right). \tag{1.85}$$

Applying integration by parts gives

$$\int_{-\infty}^{\infty} z^2\exp\left(-\frac{z^2}{2\sigma^2}\right)\,dz = -\sigma^2\left[z\exp\left(-\frac{z^2}{2\sigma^2}\right)\right]_{-\infty}^{\infty} + \sigma^2\int_{-\infty}^{\infty}\exp\left(-\frac{z^2}{2\sigma^2}\right)\,dz. \tag{1.86}$$

The boundary term is

$$\left[z\exp\left(-\frac{z^2}{2\sigma^2}\right)\right]_{-\infty}^{\infty} = \lim_{z\to\infty}\frac{z}{\exp\left(\frac{z^2}{2\sigma^2}\right)} - \lim_{z\to-\infty}\frac{z}{\exp\left(\frac{z^2}{2\sigma^2}\right)}. \tag{1.87}$$

Applying L'Hôpital's rule gives

$$\lim_{z\to\pm\infty}\frac{z}{\exp\left(\frac{z^2}{2\sigma^2}\right)} = \lim_{z\to\pm\infty}\frac{1}{\frac{z}{\sigma^2}\exp\left(\frac{z^2}{2\sigma^2}\right)} = 0. \tag{1.88}$$

Therefore,

$$\left[z\exp\left(-\frac{z^2}{2\sigma^2}\right)\right]_{-\infty}^{\infty} = 0. \tag{1.89}$$

Thus, the boundary term vanishes. Consequently,

$$\int_{-\infty}^{\infty} z^2\,\mathcal{N}(z;0,\sigma^2)\,dz = \sigma^2\int_{-\infty}^{\infty}\mathcal{N}(z;0,\sigma^2)\,dz = \sigma^2. \tag{1.90}$$

It follows that

$$\mathbb{E}[x^2] = \sigma^2 + 2\mu(0) + \mu^2(1) = \mu^2 + \sigma^2. \tag{1.91}$$

Using $\mathrm{var}[x] = \mathbb{E}[x^2] - \mathbb{E}[x]^2$, we obtain

$$\mathrm{var}[x] = \sigma^2, \tag{1.92}$$

which is why σ^2 is called the variance. The Gaussian is symmetric about μ, and its mode coincides with its mean,

$$x_{\text{mode}} = \mu. \tag{1.93}$$

For a D-dimensional random vector $\mathbf{x}$, the corresponding multivariate Gaussian distribution is

$$\mathcal{N}(\mathbf{x};\boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2}|\boldsymbol{\Sigma}|^{1/2}}\exp\left[-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right], \tag{1.94}$$

where $\boldsymbol{\mu}\in\mathbb{R}^D$ is the mean vector, $\boldsymbol{\Sigma}\in\mathbb{R}^{D\times D}$ is the covariance matrix, and $|\boldsymbol{\Sigma}|$ is the determinant of Σ. The properties of multivariate Gaussian distribution will be studied in Chapter 2.

Now suppose we observe a data set $\mathbf{x} = (x_1,\ldots,x_N)^{\mathrm{T}}$ consisting of N scalar observations drawn independently from the same Gaussian distribution with unknown parameters μ and σ^2. The observations are therefore assumed to be *independent and identically distributed* (i.i.d.),

$$x_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu,\sigma^2). \tag{1.95}$$

Independence allows the joint probability of the data set to factorize,

$$p(\mathbf{x};\mu,\sigma^2) = \prod_{n=1}^{N}\mathcal{N}(x_n;\mu,\sigma^2). \tag{1.96}$$

When the observed data $\mathbf{x}$ are held fixed and this expression is viewed as a function of μ and σ^2, it becomes the *likelihood function* for the Gaussian parameters. Instead of finding the parameter values that directly maximize the likelihood function, it is usually more convenient to maximize its logarithm. Taking the logarithm converts the product over observations into a sum,

$$\ln p(\mathbf{x}\mid\mu,\sigma^2) = \sum_{n=1}^{N}\ln\mathcal{N}(x_n\mid\mu,\sigma^2) \tag{1.97}$$

$$= -\frac{1}{2\sigma^2}\sum_{n=1}^{N}(x_n-\mu)^2 - \frac{N}{2}\ln\sigma^2 - \frac{N}{2}\ln(2\pi). \tag{1.98}$$

The log-likelihood is also numerically more stable because multiplying many small probabilities can lead to numerical underflow, whereas taking logarithms replaces products by sums. The maximum likelihood estimate of the mean is obtained by differentiating the log-likelihood with respect to μ and setting the result to zero,

$$\frac{\partial}{\partial\mu}\ln p(\mathbf{x}\mid\mu,\sigma^2) = \frac{1}{\sigma^2}\sum_{n=1}^{N}(x_n-\mu) = 0. \tag{1.99}$$

Solving for μ gives

$$\mu_{\text{ML}} = \frac{1}{N}\sum_{n=1}^{N}x_n, \tag{1.100}$$

which is simply the sample mean.

Differentiating with respect to σ^2 gives

$$\frac{\partial}{\partial \sigma^2} \ln p(\mathbf{x}; \mu, \sigma^2) = \frac{1}{2\sigma^4} \sum_{n=1}^{N} (x_n - \mu)^2 - \frac{N}{2\sigma^2}. \tag{1.101}$$

Setting the derivative equal to zero and multiply both sides by $2\sigma^4$ gives

$$\sum_{n=1}^{N} (x_n - \mu)^2 - N\sigma^2 = 0. \tag{1.102}$$

Evaluating this expression at the maximum likelihood estimate $\mu = \mu_{\mathrm{ML}}$ yields

$$\sigma^2_{\mathrm{ML}} = \frac{1}{N} \sum_{n=1}^{N} (x_n - \mu_{\mathrm{ML}})^2. \tag{1.103}$$

From Eq. 1.100 and Eq. 1.103, it follows that, for Gaussian observations, maximum likelihood estimation produces the sample mean and the sample variance (measured relative to the sample mean).

An important issue is that an estimator is itself a random variable because its value depends on the random data set from which it is computed. If repeated data sets are drawn independently from the same distribution, then μ_{ML} and σ^2_{ML} will generally take different values for different data sets. We can therefore study their expectations over all possible data sets.

For the maximum likelihood estimate of the mean μ_{ML} from Eq. 1.100, we take the expectation and using the linearity of expectation,

$$\mathbb{E}[\mu_{\mathrm{ML}}] = \mathbb{E}\left[\frac{1}{N} \sum_{n=1}^{N} x_n\right] = \frac{1}{N} \sum_{n=1}^{N} \mathbb{E}[x_n] = \frac{1}{N} \sum_{n=1}^{N} \mu = \mu, \tag{1.104}$$

so μ_{ML} is an unbiased estimator of the true mean.

To derive the expectation of the variance estimator from Eq. 1.103, we first observe that

$$\begin{aligned} \sigma^2_{\mathrm{ML}} &= \frac{1}{N} \sum_{n=1}^{N} (x_n - \mu_{\mathrm{ML}})^2 = \frac{1}{N} \sum_{n=1}^{N} [(x_n - \mu) - (\mu_{\mathrm{ML}} - \mu)]^2 \\ &= \frac{1}{N} \sum_{n=1}^{N} (x_n - \mu)^2 - \frac{2}{N} (\mu_{\mathrm{ML}} - \mu) \sum_{n=1}^{N} (x_n - \mu) + (\mu_{\mathrm{ML}} - \mu)^2. \end{aligned} \tag{1.105}$$

Since

$$\sum_{n=1}^{N} (x_n - \mu) = N(\mu_{\mathrm{ML}} - \mu), \tag{1.106}$$

we obtain

$$\sigma^2_{\mathrm{ML}} = \frac{1}{N} \sum_{n=1}^{N} (x_n - \mu)^2 - (\mu_{\mathrm{ML}} - \mu)^2. \tag{1.107}$$

Taking the expectation of both sides,

$$\mathbb{E}\left[\sigma^2_{\mathrm{ML}}\right] = \frac{1}{N} \sum_{n=1}^{N} \mathbb{E}\left[(x_n - \mu)^2\right] - \mathbb{E}\left[(\mu_{\mathrm{ML}} - \mu)^2\right]. \tag{1.108}$$

Because each x_n has variance σ^2, we have $\mathbb{E}\left[(x_n - \mu)^2\right] = \sigma^2$. Also, because we already know that μ_{ML} is unbiased, i.e., $\mathbb{E}\left[\mu_{\mathrm{ML}}\right] = \mu$, and the data points are assumed to be independent, i.e., $\mathrm{cov}(x_i, x_j), \forall i \neq i$, we obtain

$$\begin{aligned} \mathbb{E}\left[(\mu_{\mathrm{ML}} - \mu)^2\right] &= \mathrm{var}[\mu_{\mathrm{ML}}] = \mathrm{var}\left[\frac{1}{N} \sum_{n=1}^{N} x_n\right] \\ &= \frac{1}{N^2} \sum_{n=1}^{N} \mathrm{var}[x_n] = \frac{1}{N^2} N\sigma^2 = \frac{\sigma^2}{N}. \end{aligned} \tag{1.109}$$

Therefore,

$$\mathbb{E}\left[\sigma^2_{\mathrm{ML}}\right] = \sigma^2 - \frac{\sigma^2}{N} = \frac{(N-1)}{N}\sigma^2. \tag{1.110}$$

Thus, the maximum likelihood variance estimator systematically underestimates the true variance by a factor of $(N-1)/N$.

It follows from Eq. 1.110 that the following estimate for the variance parameter is unbiased

$$\widetilde{\sigma}^2 = \frac{N}{N-1}\sigma^2_{\mathrm{ML}} = \frac{1}{N-1} \sum_{n=1}^{N} (x_n - \mu_{\mathrm{ML}})^2. \tag{1.111}$$

In chapter 10, we shall see how this result arises automatically when we adopt a Bayesian approach.

Note that the bias of the maximum likelihood solution becomes less significant as the number N of data points increases, and in the limit $N \to \infty$ the maximum likelihood solution for the variance equals the true variance of the distribution that generated the data. In practice, for anything other than small N , this bias will not prove to be a serious problem. However, in more complex models with many parameters, the bias problems associated with maximum likelihood will be much more severe.

1.6 Maximum likelihood and maximum a posterior (MAP): a polynomial regression example

A polynomial curve-fitting problem can be formulated probabilistically by treating the target value t as a random variable whose distribution depends on the input x and the polynomial parameters $\mathbf{w}$. Let

$$y(x, \mathbf{w}) = \sum_{j=0}^{M} w_j x^j \tag{1.112}$$

denote the polynomial prediction. We assume that, for a given x, the observed target t is generated from a Gaussian distribution centered at $y(x, \mathbf{w})$,

$$p(t \mid x, \mathbf{w}, \beta) = \mathcal{N}\left(t; y(x, \mathbf{w}), \beta^{-1}\right), \tag{1.113}$$

where β is the precision (inverse of variance) of the observation noise. Equivalently, this model can be written as

$$t = y(x, \mathbf{w}) + \epsilon, \tag{1.114}$$

with

$$\epsilon \sim \mathcal{N}(0; \beta^{-1}). \tag{1.115}$$

Thus, the polynomial determines the conditional mean of the target, and β controls the prediction uncertainty.

Consider a training set

$$\mathcal{D} = \{(x_n, t_n)\}_{n=1}^{N}. \tag{1.116}$$

Assuming that the target observations are conditionally independent given the inputs and model parameters, the likelihood of the complete target vector $\mathbf{t} = (t_1, \ldots, t_N)^{\mathrm{T}}$ factorizes as

$$p(\mathbf{t} \mid \mathbf{x}, \mathbf{w}, \beta) = \prod_{n=1}^{N} p(t_n \mid x_n, \mathbf{w}, \beta) = \prod_{n=1}^{N} \mathcal{N}\left(t_n; y(x_n, \mathbf{w}), \beta^{-1}\right). \tag{1.117}$$

However

$$\begin{aligned} \mathcal{N}\left(t_n; y(x_n, \mathbf{w}), \beta^{-1}\right) &= \frac{1}{\sqrt{2\pi\beta^{-1}}} \exp\left[-\frac{(t_n - y(x_n, \mathbf{w}))^2}{2\beta^{-1}}\right] \\ &= \left(\frac{\beta}{2\pi}\right)^{1/2} \exp\left[-\frac{\beta}{2}(t_n - y(x_n, \mathbf{w}))^2\right]. \end{aligned} \tag{1.118}$$

Hence,

$$p(\mathbf{t} \mid \mathbf{x}, \mathbf{w}, \beta) = \left(\frac{\beta}{2\pi}\right)^{N/2} \exp\left[-\frac{\beta}{2}\sum_{n=1}^{N}(t_n - y(x_n, \mathbf{w}))^2\right]. \tag{1.119}$$

We now take the logarithm

$$\ln p(\mathbf{t} \mid \mathbf{x}, \mathbf{w}, \beta) = -\frac{\beta}{2}\sum_{n=1}^{N}(y(x_n, \mathbf{w}) - t_n)^2 + \frac{N}{2}\ln\beta - \frac{N}{2}\ln(2\pi). \tag{1.120}$$

To estimate $\mathbf{w}$ by maximum likelihood, β can be regarded as fixed. Because the last two terms do not depend on $\mathbf{w}$, and the factor β does not change the location of the optimum, the maximum likelihood problem

$$\mathbf{w}_{\mathrm{ML}} = \arg\max_{\mathbf{w}} p(\mathbf{t} \mid \mathbf{x}, \mathbf{w}, \beta), \tag{1.121}$$

is equivalent to

$$\mathbf{w}_{\mathrm{ML}} = \arg\min_{\mathbf{w}} \frac{1}{2}\sum_{n=1}^{N}(y(x_n, \mathbf{w}) - t_n)^2. \tag{1.122}$$

Hence, the familiar sum-of-squares error function arises directly from maximum likelihood estimation under the assumption of Gaussian observation noise. This illustrates the more general connection between probabilistic modeling and loss functions: the assumed observation distribution determines the form of the corresponding negative log-likelihood.

The precision parameter β can also be estimated by maximum likelihood. Differentiating the log-likelihood with respect to β gives

$$\frac{\partial}{\partial\beta}\ln p(\mathbf{t} \mid \mathbf{x}, \mathbf{w}, \beta) = -\frac{1}{2}\sum_{n=1}^{N}(y(x_n, \mathbf{w}) - t_n)^2 + \frac{N}{2\beta}. \tag{1.123}$$

Setting this derivative equal to zero and evaluating at $\mathbf{w} = \mathbf{w}_{\mathrm{ML}}$ gives

$$\frac{1}{\beta_{\mathrm{ML}}} = \frac{1}{N}\sum_{n=1}^{N}(y(x_n, \mathbf{w}_{\mathrm{ML}}) - t_n)^2. \tag{1.124}$$

Thus, the maximum likelihood estimate of the uncertainty is the mean squared residual of the fitted model.

Once $\mathbf{w}_{\mathrm{ML}}$ and β_{ML} have been estimated, prediction for a new input x is no longer represented only by the point estimate $y(x, \mathbf{w}_{\mathrm{ML}})$. Instead, the probabilistic model provides the predictive distribution

$$p(t \mid x, \mathbf{w}_{\mathrm{ML}}, \beta_{\mathrm{ML}}) = \mathcal{N}\left(t; y(x, \mathbf{w}_{\mathrm{ML}}), \beta_{\mathrm{ML}}^{-1}\right). \tag{1.125}$$

Its mean gives the usual polynomial prediction, and its variance represents uncertainty arising from the assumed observation noise. At this stage, however, maximum likelihood remains a frequentist method, so the parameters themselves are still represented by point estimates rather than by probability distributions.

A Bayesian treatment introduces a prior distribution over the polynomial coefficients. A common choice is an isotropic zero-mean Gaussian prior,

$$p(\mathbf{w} \mid \alpha) = \mathcal{N}\left(\mathbf{w}; \mathbf{0}, \alpha^{-1}\mathbf{I}\right), \tag{1.126}$$

where α is the prior precision. For an Mth-order polynomial, the parameter vector $\mathbf{w}$ contains $M+1$ coefficients, so in the multivariate Gaussian distribution given in Eq. 1.94 we have $D = M+1$, $\boldsymbol{\mu} = \mathbf{0}$, and $\boldsymbol{\Sigma} = \alpha^{-1}\mathbf{I}$. Also, because $\boldsymbol{\Sigma}$ is an $(M+1)\times(M+1)$ diagonal matrix with every diagonal entry equal to α^{-1}, its determinant is

$$|\boldsymbol{\Sigma}| = \left|\alpha^{-1}\mathbf{I}\right| = \alpha^{-(M+1)}, \tag{1.127}$$

and therefore

$$|\boldsymbol{\Sigma}|^{1/2} = \alpha^{-(M+1)/2}. \tag{1.128}$$

Its inverse is

$$\boldsymbol{\Sigma}^{-1} = (\alpha^{-1}\mathbf{I})^{-1} = \alpha\mathbf{I}. \tag{1.129}$$

Substituting these expressions into Eq. 1.94 gives

$$\begin{aligned} p(\mathbf{w} \mid \alpha) &= \frac{1}{(2\pi)^{D/2}|\mathbf{\Sigma}|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}}\mathbf{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right] \\ &= \frac{1}{(2\pi)^{(M+1)/2}\alpha^{-(M+1)/2}} \exp\left(-\frac{1}{2}\mathbf{w}^{\mathrm{T}}(\alpha\mathbf{I})\mathbf{w}\right) \\ &= \left(\frac{\alpha}{2\pi}\right)^{(M+1)/2} \exp\left(-\frac{\alpha}{2}\mathbf{w}^{\mathrm{T}}\mathbf{w}\right). \end{aligned} \tag{1.130}$$

As $p(\mathbf{w} \mid \alpha) \uparrow\downarrow \mathbf{w}^{\mathrm{T}}\mathbf{w}$, the prior $p(\mathbf{w} \mid \alpha)$ favors parameter vectors with smaller Euclidean norm and assigns progressively lower probability to vectors containing large coefficients. The quantities α and β, which control the prior and observation distributions, are examples of *hyperparameters*.

Using Bayes' theorem, the posterior distribution over $\mathbf{w}$ is

$$p(\mathbf{w} \mid \mathbf{t}, \mathbf{x}, \alpha, \beta) = \frac{p(\mathbf{t} \mid \mathbf{w}, \mathbf{x}, \alpha, \beta)p(\mathbf{w} \mid \mathbf{x}, \alpha, \beta)}{p(\mathbf{t} \mid \mathbf{x}, \alpha, \beta)}. \tag{1.131}$$

Because the prior over $\mathbf{w}$ depends only on α,

$$p(\mathbf{w} \mid \mathbf{x}, \alpha, \beta) = p(\mathbf{w} \mid \alpha), \tag{1.132}$$

and because the likelihood depends on β and $\mathbf{w}$ but not directly on α,

$$p(\mathbf{t} \mid \mathbf{x}, \mathbf{w}, \alpha, \beta) = p(\mathbf{t} \mid \mathbf{x}, \mathbf{w}, \beta), \tag{1.133}$$

and because the denominator is the evidence, obtained by marginalizing over all possible values of $\mathbf{w}$,

$$p(\mathbf{t} \mid \mathbf{x}, \alpha, \beta) = \int p(\mathbf{t} \mid \mathbf{x}, \mathbf{w}, \beta)p(\mathbf{w} \mid \alpha)\, d\mathbf{w}. \tag{1.134}$$

Therefore,

$$p(\mathbf{w} \mid \mathbf{x}, \mathbf{t}, \alpha, \beta) = \frac{p(\mathbf{t} \mid \mathbf{x}, \mathbf{w}, \beta)p(\mathbf{w} \mid \alpha)}{\int p(\mathbf{t} \mid \mathbf{x}, \mathbf{w}, \beta)p(\mathbf{w} \mid \alpha)\, d\mathbf{w}}. \tag{1.135}$$

For determining the value of $\mathbf{w}$ that maximizes the posterior, the evidence in the denominator can be ignored because it does not depend on $\mathbf{w}$. Therefore,

$$p(\mathbf{w} \mid \mathbf{x}, \mathbf{t}, \alpha, \beta) \propto p(\mathbf{t} \mid \mathbf{x}, \mathbf{w}, \beta)p(\mathbf{w} \mid \alpha). \tag{1.136}$$

The *maximum a posteriori (MAP)* estimate is then

$$\begin{aligned} \mathbf{w}_{\mathrm{MAP}} &= \arg\max_{\mathbf{w}} p(\mathbf{w} \mid \mathbf{x}, \mathbf{t}, \alpha, \beta) \\ &= \arg\max_{\mathbf{w}} \left[p(\mathbf{t} \mid \mathbf{x}, \mathbf{w}, \beta)p(\mathbf{w} \mid \alpha)\right]. \end{aligned} \tag{1.137}$$

Taking the negative logarithm,

$$\mathbf{w}_{\mathrm{MAP}} = \arg\min_{\mathbf{w}} \left[-\ln p(\mathbf{t} \mid \mathbf{x}, \mathbf{w}, \beta) - \ln p(\mathbf{w} \mid \alpha)\right]. \tag{1.138}$$

Combining with Eq. 1.120 and Eq. 1.130

$$\begin{aligned} \mathbf{w}_{\mathrm{MAP}} = \arg\min_{\mathbf{w}} \Bigg[&\frac{\beta}{2}\sum_{n=1}^{N}(y(x_n, \mathbf{w}) - t_n)^2 - \frac{N}{2}\ln\beta + \frac{N}{2}\ln(2\pi) \\ &+ \frac{\alpha}{2}\mathbf{w}^{\mathrm{T}}\mathbf{w} - \frac{M+1}{2}\ln\alpha + \frac{M+1}{2}\ln(2\pi)\Bigg]. \end{aligned} \tag{1.139}$$

Discarding terms that do not depend on $\mathbf{w}$ and dividing the objective by a positive constant $\beta/2$ do not change the optimum. Therefore, the MAP estimate is equivalently given by

$$\mathbf{w}_{\mathrm{MAP}} = \arg\min_{\mathbf{w}} \left[\sum_{n=1}^{N}(y(x_n, \mathbf{w}) - t_n)^2 + \lambda\mathbf{w}^{\mathrm{T}}\mathbf{w}\right]. \tag{1.140}$$

where $\lambda = \alpha/\beta$. Thus, maximizing the posterior distribution is equivalent to minimizing a sum-of-squares error with an L_2 regularization term, commonly known as *ridge regularization*.

This result is not specific to polynomial curve fitting. More generally, whenever the observation errors are assumed to be independent Gaussian with common variance β^{-1} and the model parameters are assigned a zero-mean Gaussian prior with variance α^{-1}, MAP estimation leads to an objective consisting of a squared-error term together with an L_2 regularization term, with regularization coefficient $\lambda = \beta^{-1}/\alpha^{-1}$. The Gaussian observation model produces the squared-error term through its negative log-likelihood, and the Gaussian parameter prior produces the L_2 penalty.

Although MAP is a Bayesian approach because it introduces a prior distribution $p(\mathbf{w} \mid \alpha)$ for the parameters, it still reduces the uncertainty in $\mathbf{w}$ to a single point estimate $\mathbf{w}_{\mathrm{MAP}}$. The next section introduces a full Bayesian approach.

1.7 Bayesian curve fitting

A fully Bayesian treatment differs from maximum likelihood or MAP estimation in that the entire posterior distribution $p(\mathbf{w} \mid \mathbf{x}, \mathbf{t})$ is retained and marginalized when making predictions. In the curve-fitting problem, given training inputs $\mathbf{x} = (x_1, \ldots, x_N)^{\mathrm{T}}$, corresponding targets $\mathbf{t} = (t_1, \ldots, t_N)^{\mathrm{T}}$, and a new input x, the goal is to determine the predictive distribution of the corresponding target t.

Instead of predicting with a single parameter vector such as $\mathbf{w}_{\mathrm{ML}}$ or $\mathbf{w}_{\mathrm{MAP}}$, the predictive distribution is obtained by averaging the predictions associated with all possible parameter values, weighted by their posterior probability.

$$p(t \mid x, \mathbf{x}, \mathbf{t}) = \int p(t \mid x, \mathbf{w})p(\mathbf{w} \mid \mathbf{x}, \mathbf{t})\, d\mathbf{w}. \tag{1.141}$$

For polynomial curve fitting, it is convenient to write the model using the feature vector $\boldsymbol{\phi}(x)$ with elements $\phi_i(x) = x^i$ for $i = 0, \ldots, M$. The Gaussian model now becomes

$$p(t \mid x, \mathbf{w}, \beta) = \mathcal{N}\left(t; \mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(x), \beta^{-1}\right), \tag{1.142}$$

For fixed training data $\mathbf{x}$ and $\mathbf{t}$, the model in Eq. 1.142 is viewed as a function of $\mathbf{w}$ and β, giving the likelihood.

$$p(\mathbf{t} \mid \mathbf{x}, \mathbf{w}, \beta) = \prod_{n=1}^{N} \mathcal{N}\left(t_n; \mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(x_n), \beta^{-1}\right). \tag{1.143}$$

The Gaussian prior is

$$p(\mathbf{w} \mid \alpha) = \mathcal{N}\left(\mathbf{w}; \mathbf{0}, \alpha^{-1}\mathbf{I}\right). \tag{1.144}$$

Similar to Eq. 1.119 and Eq. 1.130, the likelihood and the prior can be written, up to factors independent of $\mathbf{w}$, as

$$p(\mathbf{t} \mid \mathbf{x}, \mathbf{w}, \beta) \propto \exp\left[-\frac{\beta}{2}\sum_{n=1}^{N}\left(t_n - \mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(x_n)\right)^2\right] \tag{1.145}$$

and

$$p(\mathbf{w} \mid \alpha) \propto \exp\left(-\frac{\alpha}{2}\mathbf{w}^{\mathrm{T}}\mathbf{w}\right). \tag{1.146}$$

The posterior is proportional to the product of the likelihood and the prior (Eq. 1.136), therefore

$$p(\mathbf{w} \mid \mathbf{x}, \mathbf{t}, \alpha, \beta) \propto \exp\left[-\frac{1}{2}\left\{\beta\sum_{n=1}^{N}\left(t_n - \mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(x_n)\right)^2 + \alpha\mathbf{w}^{\mathrm{T}}\mathbf{w}\right\}\right]. \tag{1.147}$$

Expand the squared residual,

$$\left(t_n - \mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(x_n)\right)^2 = t_n^2 - 2t_n\mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(x_n) + \mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(x_n)\boldsymbol{\phi}(x_n)^{\mathrm{T}}\mathbf{w}. \tag{1.148}$$

Summing over all observations and multiplyig with β give

$$\beta\sum_{n=1}^{N}\left(t_n - \mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(x_n)\right)^2 = \beta\sum_{n=1}^{N}t_n^2 - 2\beta\mathbf{w}^{\mathrm{T}}\sum_{n=1}^{N}\boldsymbol{\phi}(x_n)t_n + \beta\mathbf{w}^{\mathrm{T}}\left[\sum_{n=1}^{N}\boldsymbol{\phi}(x_n)\boldsymbol{\phi}(x_n)^{\mathrm{T}}\right]\mathbf{w}. \tag{1.149}$$

Substituting this expansion into the posterior exponent and dropping the term $\beta\sum_{n=1}^{N}t_n^2$ that does not depend on $\mathbf{w}$ gives

$$p(\mathbf{w} \mid \mathbf{x}, \mathbf{t}, \alpha, \beta) \propto \exp\left[-\frac{1}{2}\left\{\mathbf{w}^{\mathrm{T}}\left(\alpha\mathbf{I} + \beta\sum_{n=1}^{N}\boldsymbol{\phi}(x_n)\boldsymbol{\phi}(x_n)^{\mathrm{T}}\right)\mathbf{w} - 2\beta\mathbf{w}^{\mathrm{T}}\sum_{n=1}^{N}\boldsymbol{\phi}(x_n)t_n\right\}\right]. \tag{1.150}$$

Define

$$\mathbf{S}^{-1} = \alpha\mathbf{I} + \beta\sum_{n=1}^{N}\boldsymbol{\phi}(x_n)\boldsymbol{\phi}(x_n)^{\mathrm{T}}. \tag{1.151}$$

At this point, the posterior has the form

$$p(\mathbf{w} \mid \mathbf{x}, \mathbf{t}, \alpha, \beta) \propto \exp\left[-\frac{1}{2}\left(\mathbf{w}^{\mathrm{T}}\mathbf{S}^{-1}\mathbf{w} - 2\beta\mathbf{w}^{\mathrm{T}}\sum_{n=1}^{N}\boldsymbol{\phi}(x_n)t_n\right)\right]. \tag{1.152}$$

We now want to rewrite the expression inside the exponential as a quadratic centered at some vector $\mathbf{m}_N$. We introduce $\mathbf{m}_N$ as the vector that will become the center, and therefore the mean, of the posterior Gaussian. A quadratic centered at $\mathbf{m}_N$ has the form

$$(\mathbf{w} - \mathbf{m}_N)^{\mathrm{T}}\mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}_N). \tag{1.153}$$

Expanding this expression gives

$$(\mathbf{w} - \mathbf{m}_N)^{\mathrm{T}}\mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}_N) = \mathbf{w}^{\mathrm{T}}\mathbf{S}^{-1}\mathbf{w} - \mathbf{w}^{\mathrm{T}}\mathbf{S}^{-1}\mathbf{m}_N - \mathbf{m}_N^{\mathrm{T}}\mathbf{S}^{-1}\mathbf{w} + \mathbf{m}_N^{\mathrm{T}}\mathbf{S}^{-1}\mathbf{m}_N. \tag{1.154}$$

Because $\mathbf{m}_N^{\mathrm{T}}\mathbf{S}^{-1}\mathbf{w}$ is a scalar, its tranpose is equal to itself, and because $\mathbf{S}^{-1}$ is symmetric,

$$\mathbf{m}_N^{\mathrm{T}}\mathbf{S}^{-1}\mathbf{w} = \mathbf{w}^{\mathrm{T}}\mathbf{S}^{-1}\mathbf{m}_N, \tag{1.155}$$

so

$$(\mathbf{w} - \mathbf{m}_N)^{\mathrm{T}}\mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}_N) = \mathbf{w}^{\mathrm{T}}\mathbf{S}^{-1}\mathbf{w} - 2\mathbf{w}^{\mathrm{T}}\mathbf{S}^{-1}\mathbf{m}_N + \mathbf{m}_N^{\mathrm{T}}\mathbf{S}^{-1}\mathbf{m}_N. \tag{1.156}$$

We now choose $\mathbf{m}_N$ so that the linear term in $\mathbf{w}$ matches the linear term in the posterior. To let

$$-2\mathbf{w}^{\mathrm{T}}\mathbf{S}^{-1}\mathbf{m}_N = -2\beta\mathbf{w}^{\mathrm{T}}\sum_{n=1}^{N}\boldsymbol{\phi}(x_n)t_n, \tag{1.157}$$

we require

$$\mathbf{S}^{-1}\mathbf{m}_N = \beta\sum_{n=1}^{N}\boldsymbol{\phi}(x_n)t_n. \tag{1.158}$$

Multiplying both sides by $\mathbf{S}$ gives

$$\mathbf{m}_N = \beta\mathbf{S}\sum_{n=1}^{N}\boldsymbol{\phi}(x_n)t_n. \tag{1.159}$$

Thus, $\mathbf{m}_N$ is chosen precisely so that the original quadratic expression in $\mathbf{w}$ can be written as a quadratic centered at $\mathbf{m}_N$. Using this choice,

$$\mathbf{w}^{\mathrm{T}}\mathbf{S}^{-1}\mathbf{w} - 2\beta\mathbf{w}^{\mathrm{T}}\sum_{n=1}^{N}\boldsymbol{\phi}(x_n)t_n = (\mathbf{w} - \mathbf{m}_N)^{\mathrm{T}}\mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}_N) - \mathbf{m}_N^{\mathrm{T}}\mathbf{S}^{-1}\mathbf{m}_N. \tag{1.160}$$

The final term does not depend on $\mathbf{w}$, so it is absorbed into the normalization constant. Therefore,

$$p(\mathbf{w} \mid \mathbf{x}, \mathbf{t}, \alpha, \beta) \propto \exp\left[-\frac{1}{2}(\mathbf{w}-\mathbf{m}_N)^{\mathrm{T}}\mathbf{S}^{-1}(\mathbf{w}-\mathbf{m}_N)\right]. \tag{1.161}$$

This is the form of a multivariate Gaussian centered at $\mathbf{m}_N$. Hence,

$$p(\mathbf{w} \mid \mathbf{x}, \mathbf{t}, \alpha, \beta) = \mathcal{N}(\mathbf{w}; \mathbf{m}_N, \mathbf{S}), \tag{1.162}$$

so $\mathbf{m}_N$ is the posterior mean and $\mathbf{S}$ is the posterior covariance.

Substituting the Gaussian observation model and Gaussian posterior into the predictive distribution in Eq. 1.141 gives

$$p(t \mid x, \mathbf{x}, \mathbf{t}) = \int \mathcal{N}\left(t; \mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(x), \beta^{-1}\right) \mathcal{N}(\mathbf{w}; \mathbf{m}_N, \mathbf{S})\, d\mathbf{w}, \tag{1.163}$$

For a fixed input x, the feature vector $\boldsymbol{\phi}(x)$ is fixed. Because the posterior distribution of $\mathbf{w}$ is a Gaussian given in Eq. 1.162, a linear transformation of $\mathbf{w}$ is also a Gaussian,

$$p(\mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(x) \mid \mathbf{x}, \mathbf{t}, \alpha, \beta) = \mathcal{N}\left(\mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(x); \mathbf{m}_N^{\mathrm{T}}\boldsymbol{\phi}(x), \boldsymbol{\phi}(x)^{\mathrm{T}}\mathbf{S}\boldsymbol{\phi}(x)\right). \tag{1.164}$$

To see where these parameters come from, the mean of the transformed variable is

$$\mathbb{E}_{\mathbf{w}}\left[\mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(x)\right] = \mathbb{E}_{\mathbf{w}}[\mathbf{w}]^{\mathrm{T}}\boldsymbol{\phi}(x) = \mathbf{m}_N^{\mathrm{T}}\boldsymbol{\phi}(x), \tag{1.165}$$

and its variance is

$$\mathrm{var}_{\mathbf{w}}\left[\mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(x)\right] = \boldsymbol{\phi}(x)^{\mathrm{T}}\,\mathrm{cov}[\mathbf{w}]\boldsymbol{\phi}(x) = \boldsymbol{\phi}(x)^{\mathrm{T}}\mathbf{S}\boldsymbol{\phi}(x). \tag{1.166}$$

When we rewrite the Gaussian model in Eq. 1.142 as

$$t(x) = \mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(x) + \epsilon, \qquad \text{with} \quad \epsilon \sim \mathcal{N}(0, \beta^{-1}), \tag{1.167}$$

We see that $t(x)$ is the sum of two independent Gaussian random variables, with the distribution of $\mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(x)$ given in Eq. 1.164. Therefore, $t(x)$ is itself Gaussian.

The predictive mean is:

$$m(x) = \mathbb{E}[t(x)] = \mathbf{m}_N^{\mathrm{T}}\boldsymbol{\phi}(x) + 0 = \beta\boldsymbol{\phi}(x)^{\mathrm{T}}\mathbf{S}\sum_{n=1}^{N}\boldsymbol{\phi}(x_n)t_n. \tag{1.168}$$

Similarly, because the two terms are independent, their variances add,

$$s^2(x) = \mathrm{var}[t(x)] = \mathrm{var}\left[\mathbf{w}^{\mathrm{T}}\boldsymbol{\phi}(x)\right] + \mathrm{var}[\epsilon] = \boldsymbol{\phi}(x)^{\mathrm{T}}\mathbf{S}\boldsymbol{\phi}(x) + \beta^{-1}. \tag{1.169}$$

Therefore, the predictive distribution becomes

$$p(t \mid x, \mathbf{x}, \mathbf{t}) = \mathcal{N}\left(t; m(x), s^2(x)\right), \tag{1.170}$$

where $m(x)$ and $s^2(x)$ are given in Eq. 1.168 and Eq. 1.169

The predictive variance has two distinct sources,

$$s^2(x) = \underbrace{\boldsymbol{\phi}(x)^{\mathrm{T}}\mathbf{S}\boldsymbol{\phi}(x)}_{\text{parameter uncertainty}} + \underbrace{\beta^{-1}}_{\text{observation noise}}. \tag{1.171}$$

This decomposition is a key distinction between point-estimate and fully Bayesian prediction. A maximum likelihood prediction with fixed $\mathbf{w}_{\mathrm{ML}}$ accounts only for observation noise, whereas the Bayesian predictive distribution additionally accounts for uncertainty in the model parameters. As more informative training data are observed, the posterior covariance $\mathbf{S}$ of the parameter typically decreases, reducing the contribution from parameter uncertainty while the irreducible observation-noise variance β^{-1} remains.

1.8 Model Selection

Model selection concerns choosing the model class and its effective complexity. Because training error is optimistically biased and generally decreases as model flexibility increases, model selection must be based on data not used to fit the candidate model.

When data is scarce, a common procedure is to first split the full data set into a *training set* and a held-out *test set*. Model selection is then performed using only the training set. With S-fold cross-validation, the training set is partitioned into S folds; for each fold, the model is fitted on $S-1$ folds and evaluated on the remaining fold. If E_s denotes the validation error on fold s, the cross-validation estimate is

$$E_{\mathrm{CV}} = \frac{1}{S}\sum_{s=1}^{S} E_s. \tag{1.172}$$

After selecting the model, it is typically refitted using the entire training set and evaluated once on the held-out test set.

The special case $S = N$, where N is the number of training observations, is *leave-one-out cross-validation*. The main disadvantage of cross-validation is computational cost since every candidate model must be trained S times.

An alternative is to augment goodness of fit with an explicit complexity penalty. For example, the Akaike information criterion can be written as

$$\mathrm{AIC} = -2\ln p(\mathcal{D} \mid \mathbf{w}_{\mathrm{ML}}) + 2M, \tag{1.173}$$

where M is the number of trainable parameters. The negative log-likelihood term rewards fit to the observed data, while the second term penalizes model complexity.

A variant of this quantity, called the *Bayesian information criterion (BIC)*, will be discussed in Chapter 4. A fully Bayesian approach to model comparison, introduced in Chapter 3, provides a more principled treatment of model selection by integrating over parameter uncertainty, so that model complexity is controlled implicitly rather than through an explicit penalty term.

1.9 The curse of dimensionality

In practical pattern-recognition problems, the input $\mathbf{x}$ is often high-dimensional. As the dimensionality D increases, the volume of the input space grows extremely rapidly, so a fixed number of training observations becomes increasingly sparse relative to the space that must be represented. This phenomenon is known as the *curse of dimensionality*.

A simple illustration is obtained by partitioning each input dimension into K intervals. In one dimension this produces K cells, in two dimensions K^2 cells, and in D dimensions

$$N_{\text{cells}} = K^D. \tag{1.174}$$

Thus, maintaining the same spatial resolution in every dimension requires a number of cells that grows exponentially with D. A classifier that assigns a test point according to the training observations lying in the same cell therefore requires an exponentially increasing amount of data merely to prevent most cells from becoming empty. Local methods based on nearby observations encounter the same underlying problem because neighborhoods become increasingly sparsely populated as D grows.

The curse of dimensionality also appears in parametric models because the number of distinct terms needed to represent interactions among the input variables can grow very rapidly with the input dimension D. Suppose that

$$\mathbf{x} = (x_1, \ldots, x_D)^{\mathrm{T}} \tag{1.175}$$

contains D input variables, and consider a polynomial model containing all terms up to total degree three,

$$y(\mathbf{x}, \mathbf{w}) = w_0 + \sum_{i=1}^{D} w_i x_i + \sum_{i=1}^{D}\sum_{j=1}^{D} w_{ij} x_i x_j + \sum_{i=1}^{D}\sum_{j=1}^{D}\sum_{k=1}^{D} w_{ijk} x_i x_j x_k. \tag{1.176}$$

The linear part contains D distinct terms,

$$x_1,\ x_2,\ \ldots,\ x_D. \tag{1.177}$$

At first sight, the quadratic double sum appears to contain D^2 terms because there are D possible values for i and D possible values for j. However, products such as $x_i x_j$ are invariant under permutations of their indices. Thus, their coefficients can be combined into a single coefficient. As a result, there are D squared terms and $\binom{D}{2}$ pairwise interaction terms, giving

$$N_2 = D + \binom{D}{2} = D + \frac{D(D-1)}{2} = \binom{D+1}{2} \tag{1.178}$$

distinct monomials of degree two.

More generally, a monomial of total degree m in D variables has the form

$$x_1^{a_1} x_2^{a_2} \cdots x_D^{a_D}, \qquad a_i \in \{0, 1, 2, \ldots, m\}. \tag{1.179}$$

subject to

$$\sum_{i=1}^{D} a_i = m. \tag{1.180}$$

The number of nonnegative integer solutions to this equation is

$$N_m = \binom{D+m-1}{m}. \tag{1.181}$$

If the polynomial contains all monomials from degree 0 through degree M, the total number of independent coefficients is therefore

$$N_{\text{coeff}} = \sum_{m=0}^{M} \binom{D+m-1}{m} = \binom{D+M}{M}. \tag{1.182}$$

For a cubic polynomial, $M = 3$, so

$$N_{\text{coeff}} = \binom{D+3}{3} = \frac{(D+3)(D+2)(D+1)}{6}. \tag{1.183}$$

Thus, even though the compact polynomial expression contains sums with D, D^2, and D^3 indexed terms, the number of coefficients is smaller. Nevertheless, the number of coefficients still grows rapidly with dimension D. When D is large compared with a fixed polynomial order M, each factor in the numerator is approximately D, giving

$$\binom{D+M}{M} \sim \frac{D^M}{M!}. \tag{1.184}$$

Hence, for a fixed polynomial order M, the number of parameters grows approximately as D^M. The growth is therefore polynomial rather than exponential in D. However, it can still become prohibitively large even for moderately high-dimensional inputs.

This exmaple is one manifestation of the curse of dimensionality. Increasing the number of input variables does not merely enlarge the input space; it can also cause the number of parameters required to represent interactions among those variables to grow combinatorially. As the number of parameters increases, more data are generally required for reliable estimation and it increases both the computational cost and the risk of overfitting.

A closely related high-dimensional effect appears even for a simple isotropic Gaussian distribution. Consider the D-dimensional standard Gaussian

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{D/2}} \exp\left(-\frac{1}{2}\mathbf{x}^{\mathrm{T}}\mathbf{x}\right), \tag{1.185}$$

where $\mathbf{x} \in \mathbb{R}^D$. The distance of $\mathbf{x}$ from the origin is

$$r = \|\mathbf{x}\| = \sqrt{\mathbf{x}^{\mathrm{T}}\mathbf{x}}. \tag{1.186}$$

The density at any individual point depends only on its distance from the origin,

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{D/2}} \exp\left(-\frac{r^2}{2}\right). \tag{1.187}$$

The density is therefore largest at $r = 0$ and decreases exponetially as r increases. However, this does not mean that most of the probability mass lies near the origin. The reason is that probability mass depends on both the density at each point and the number of points available at a given radius.

To see this, consider a thin spherical shell with radius r and thickness dr. All points in this shell have approximately the same Gaussian density because they have approximately the same value of r. The probability mass in the shell is therefore approximately

$$p_r(r)\, dr = p(\mathbf{x}) \times \text{surface area of the sphere at radius } r \times dr. \tag{1.188}$$

The surface area of a D-dimensional sphere of radius r is

$$A_D(r) = \frac{2\pi^{D/2}}{\Gamma(D/2)} r^{D-1}. \tag{1.189}$$

Therefore,

$$p_r(r) = A_D(r) \frac{1}{(2\pi)^{D/2}} \exp\left(-\frac{r^2}{2}\right) \tag{1.190}$$

$$= \frac{1}{2^{D/2-1}\Gamma(D/2)} r^{D-1} \exp\left(-\frac{r^2}{2}\right), \qquad r \geq 0. \tag{1.191}$$

The Gaussian factor $\exp(-r^2/2)$ decreases as r increases, because individual points farther from the origin have lower density. In contrast, the geometric factor r^{D-1} increases rapidly with r, because the surface area of a high-dimensional sphere increases rapidly with radius. The radial probability density $p_r(r)$ is determined by the competition between these two effects.

To find the radius at which the probability mass is most concentrated, we maximize $p_r(r)$. This is equivalent to maximizing its logarithm,

$$\ln p_r(r) = (D-1) \ln r - \frac{r^2}{2} + \text{const.} \tag{1.192}$$

Differentiating with respect to r gives

$$\frac{d}{dr} \ln p_r(r) = \frac{D-1}{r} - r. \tag{1.193}$$

Setting the derivative equal to zero gives

$$r_{\text{mode}} = \sqrt{D-1}. \tag{1.194}$$

For $D = 1$, the probability mass is concentrated most strongly at the center, $r = 0$. For $D = 2$, the largest amount of probability mass per small change in radius lies on a ring at $r = 1$. In higher dimensions, most of the probability mass is concentrated in a thin spherical shell located farther from the origin at $r = \sqrt{D-1}$. Although each individual point in this thin shell has a much lower density than the origin, the shell contains vastly more points, so it contains much more total probability mass.

The curse of dimensionality does not, however, imply that learning from high-dimensional data is necessarily impossible. The key reason is that real data often do not occupy the entire D-dimensional space. Instead, they may be concentrated near a much lower-dimensional subset. If the possible observations can be described by only d independent degrees of freedom, with $d \ll D$, then the data may lie approximately on a d-dimensional manifold embedded in $\mathbb{R}^D$. The quantity d is referred to as the *intrinsic dimensionality* of the data.

For example, an image containing D pixels can be represented as

$$\mathbf{x} = (x_1, \ldots, x_D)^{\mathrm{T}}, \tag{1.195}$$

so the image is formally a point in $\mathbb{R}^D$. However, suppose all images contain the same object and differ only in horizontal position, vertical position, and orientation. Although the pixel representation may have thousands of dimensions, the variability among the images is controlled primarily by only three quantities. The set of valid images might therefore occupy a structure with intrinsic dimension close to three within the much larger pixel space. This distinction is important because a learning algorithm need not model the entire space $\mathbb{R}^D$, but only the much smaller structure in which the data are concentrated. Modern dimensionality-reduction and representation-learning methods attempt to discover or exploit such low-dimensional structure.

A second property that can reduce the practical severity of the curse of dimensionality is *smoothness*. Suppose that a target is represented by a function $f(\mathbf{x})$. A smoothness assumption states, roughly, that nearby inputs tend to produce similar outputs,

$$\|\mathbf{x} - \mathbf{x}'\| \text{ small} \quad \Longrightarrow \quad |f(\mathbf{x}) - f(\mathbf{x}')| \text{ small}. \tag{1.196}$$

Under this assumption, observing the target at one point provides useful information about the target at nearby points. Therefore, smoothness allows nearby observations to share predictive information.

Sparsity assumes that only a relatively small subset of the available variables, features, or model parameters is needed to describe the relevant structure in the data. For example, even if an input $\mathbf{x} \in \mathbb{R}^D$ contains a very large number of features, the target may depend strongly on only a small subset of them. Similarly, a model may contain a large parameter vector $\mathbf{w}$ while only a small number of entries are substantially different from zero. By reducing the number of active variables or parameters, sparsity lowers the effective number of degrees of freedom that must be estimated from data and can improve generalization. Common mechanisms for encouraging sparsity include feature selection and regularization methods such as L_1 regularization.

Locality assumes that interactions are strongest among inputs that are close according to some appropriate notion of neigh-

borhood, while distant inputs interact weakly or not at all. This reduces the need to model arbitrary interactions among all components of a high-dimensional input. For example, in an image, nearby pixels are often much more strongly related than pixels located far apart. A model can therefore restrict each unit to operate only on a local region rather than on the entire input. If each local computation involves only k nearby variables with $k \ll D$, then the effective dimensionality of each interaction is much smaller than the full ambient dimension. Locality is exploited explicitly in methods such as convolutional neural networks, nearest-neighbor methods, and local kernel models.

Parameter sharing reduces model complexity by using the same parameters across multiple locations or parts of the input rather than learning a separate set of parameters for each one. Convolutional neural networks provide a standard example, where the same convolutional filter is repeatedly applied across the entire spatial domain. This substantially reduces the number of trainable parameters, imposes useful structural constraints on the model, and allows knowledge learned in one region of the input to generalize to other regions.

Therefore, successful learning in high-dimensional spaces generally relies on some form of structure or property in the data and/or model. It could include low intrinsic dimensionality, smoothness, sparsity, locality, parameter sharing, and other constraints that reduce the effective complexity of the learning problem relative to the nominal ambient dimension D.

1.10 Decision Theory

Probability theory provides a mathematical framework for representing and manipulating uncertainty, but in many pattern recognition problems the ultimate objective is not merely to estimate probabilities. Instead, we must use those probabilities to make a decision. Decision theory provides the framework for selecting an action when the quantities of interest are uncertain, thereby connecting probabilistic inference with prediction and decision making.

Suppose we observe an input vector $\mathbf{x}$ and wish to predict a corresponding target variable t. In regression problems, t is typically continuous, whereas in classification problems it represents a discrete class label. The joint distribution $p(\mathbf{x}, t)$ provides a complete probabilistic description of the relationship between inputs and targets. Estimating such a distribution from training data is an *inference* problem. Once the relevant probability distribution has been obtained, however, we must still determine what prediction or action should be taken for a particular input $\mathbf{x}$. This second stage is the *decision* problem.

For classification, suppose the target belongs to one of K classes $\{C_1, \ldots, C_K\}$. Given an observed input $\mathbf{x}$, Bayes' theorem gives the posterior probability of class C_k as

$$p(C_k \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid C_k)p(C_k)}{p(\mathbf{x})} = \frac{p(\mathbf{x} \mid C_k)p(C_k)}{\sum_{j=1}^{K} p(\mathbf{x} \mid C_j)p(C_j)}. \quad (1.197)$$

Here, $p(C_k)$ is the prior probability of class C_k, representing our uncertainty about the class before observing $\mathbf{x}$. The likelihood $p(\mathbf{x} \mid C_k)$ describes how probable the observed input is under class C_k. The posterior probability $p(C_k \mid \mathbf{x})$ is the probability of class C_k after incorporating the information contained in $\mathbf{x}$. The denominator normalizes the posterior probabilities so that they sum to one over all classes.

The posterior probabilities themselves do not yet specify what action should be taken. A decision rule must map the observed input, or equivalently the posterior probabilities associated with that input, to a particular action. In the simplest classification setting, where every incorrect classification is treated equally, we will show that minimizing the expected loss is equivalent to assigning $\mathbf{x}$ to the class with the largest posterior probability. In many applications, the appropriate decision rule depends not only on the posterior probabilities, but also on the loss associated with each possible error. When different misclassifications have different consequences, the optimal decision is therefore the one that minimizes the expected loss rather than simply the probability of error.

Minimizing the Misclassification Rate

We first assume that all types of errors have the same consequence. As a result, the objective of a classification problem is to minimize the probability of assigning an input to the wrong class. A decision rule assigns every possible input $\mathbf{x}$ to one of the K classes $\{C_1, \ldots, C_K\}$. This partitions the input space into *decision regions* $\{\mathcal{R}_1, \ldots, \mathcal{R}_K\}$, where every $\mathbf{x} \in \mathcal{R}_k$ is assigned to class C_k. The boundaries separating these regions are called *decision boundaries* or *decision surfaces*.

Now consider all inputs that fall in a particular region $\mathcal{R}_k$. The classifier assigns all of these inputs to class C_k. A classification error occurs whenever the true class is some C_j with $j \neq k$. For one such class C_j, the probability of this type of error is

$$p(\mathbf{x} \in \mathcal{R}_k, C_j) = \int_{\mathcal{R}_k} p(\mathbf{x}, C_j)\, d\mathbf{x}. \quad (1.198)$$

Because the true class could be any of the remaining $K - 1$ classes, the contribution of region $\mathcal{R}_k$ to the total probability of misclassification is

$$\sum_{j \neq k} \int_{\mathcal{R}_k} p(\mathbf{x}, C_j)\, d\mathbf{x} = \int_{\mathcal{R}_k} \sum_{j \neq k} p(\mathbf{x}, C_j)\, d\mathbf{x}. \quad (1.199)$$

Finally, summing this error probability over all decision regions gives the total probability of misclassification,

$$p(\text{mistake}) = \sum_{k=1}^{K} \int_{\mathcal{R}_k} \sum_{j \neq k} p(\mathbf{x}, C_j)\, d\mathbf{x}. \quad (1.200)$$

Because we place no restrictions on the form of the decision rule, the class assigned at one value of $\mathbf{x}$ can be chosen independently of the class assigned at another value of $\mathbf{x}$. The collection of these individual assignments then determines the decision regions. For a given $\mathbf{x}$, we choose the class C_k that minimize the local contribution to the misclassification probability

$$\sum_{j \neq k} p(\mathbf{x}, C_j) = p(\mathbf{x}) - p(\mathbf{x}, C_k). \tag{1.201}$$

Because $p(\mathbf{x})$ is the same for every class at a fixed $\mathbf{x}$, minimizing this error contribution is equivalent to maximizing $p(\mathbf{x}, C_k)$. Thus, each $\mathbf{x}$ should be assigned to the class with the largest joint probability,

$$\hat{C}(\mathbf{x}) = \arg\max_{C_k} p(\mathbf{x}, C_k). \tag{1.202}$$

Using the product rule,

$$p(\mathbf{x}, C_k) = p(C_k \mid \mathbf{x}) p(\mathbf{x}), \tag{1.203}$$

and again, because $p(\mathbf{x})$ does not depend on the class C_k, maximizing the joint probability $p(\mathbf{x}, C_k)$ is equivalent to maximizing the posterior probability $p(C_k \mid \mathbf{x})$. Hence, the decision rule that minimizes the misclassification probability becomes

$$\hat{C}(\mathbf{x}) = \arg\max_{C_k} p(C_k \mid \mathbf{x}). \tag{1.204}$$

The probability of making a correct classification is

$$p(\text{correct}) = \sum_{k=1}^{K} p(\mathbf{x} \in \mathcal{R}_k, C_k) = \sum_{k=1}^{K} \int_{\mathcal{R}_k} p(\mathbf{x}, C_k)\, d\mathbf{x}. \tag{1.205}$$

It can also be shown that maximizing $p(\text{correct})$ is equivalent to minimizing $p(\text{mistake})$. In both cases, the optimal decision is obtained by assigning $\mathbf{x}$ to the class with the largest posterior probability, as in Eq. 1.204. The resulting decision boundaries occur where the posterior probabilities of competing classes are equal.

Minimizing the Expected Loss

To account for different consequences associated with different types of errors, we introduce a *loss function*. Suppose that the true class is C_k and the classifier assigns the input $\mathbf{x}$ to class C_j. We denote the loss associated with this outcome by L_{kj}. The collection of these values forms a *loss matrix*, where the element L_{kj} specifies the cost of assigning an observation from the true class C_k to class C_j. Typically, $L_{kk} = 0$ if a correct classification incurs no loss, whereas the off-diagonal elements are postive and may differ depending on the consequences of different types of misclassification.

Let $\mathcal{R}_j$ denote the decision region for class C_j, so that every $\mathbf{x} \in \mathcal{R}_j$ is assigned to C_j. If the true class is C_k, the probability density associated with this outcome is $p(\mathbf{x}, C_k)$. Therefore, the expected loss contributed by assigning inputs in $\mathcal{R}_j$ to class C_j when their true class is C_k is

$$\int_{\mathcal{R}_j} L_{kj} p(\mathbf{x}, C_k)\, d\mathbf{x}. \tag{1.206}$$

Summing over all true classes and all decision regions gives the total expected loss,

$$\mathbb{E}[L] = \sum_j \sum_k \int_{\mathcal{R}_j} L_{kj} p(\mathbf{x}, C_k)\, d\mathbf{x} = \sum_j \int_{\mathcal{R}_j} \sum_k L_{kj} p(\mathbf{x}, C_k)\, d\mathbf{x}. \tag{1.207}$$

As before, if no restrictions are placed on the form of the decision rule, the class assigned at one value of $\mathbf{x}$ can be chosen independently of the class assigned at another value. Therefore, minimizing the total expected loss can be done separately for each $\mathbf{x}$. If $\mathbf{x}$ is assigned to class C_j, its local contribution to the expected loss is

$$\sum_k L_{kj} p(\mathbf{x}, C_k) = p(\mathbf{x}) \sum_k L_{kj} p(C_k \mid \mathbf{x}). \tag{1.208}$$

Because $p(\mathbf{x})$ is independent of the chosen class C_j, it does not affect which decision minimizes the loss. We therefore define the conditional expected loss associated with assigning $\mathbf{x}$ to class C_j as

$$R(C_j \mid \mathbf{x}) = \sum_k L_{kj} p(C_k \mid \mathbf{x}), \tag{1.209}$$

where $R(C_j \mid \mathbf{x})$ is also commonly called the *conditional risk*. The optimal decision rule assigns $\mathbf{x}$ to the class whose conditional expected loss is smallest,

$$\hat{C}(\mathbf{x}) = \arg\min_{C_j} \sum_k L_{kj} p(C_k \mid \mathbf{x}). \tag{1.210}$$

Thus, the posterior probabilities describe the uncertainty about the true class, and the loss matrix describes the consequences of the possible decisions. The optimal classifier combines both quantities and chooses the decision with the minimum expected loss.

We now show that the maximum-posterior classification rule in Eq. 1.204 is a special case of the more general minimum-expected-loss rule in Eq. 1.210 when all misclassification errors have the same loss. Let

$$L_{kj} = \begin{cases} 0, & k = j, \\ L, & k \neq j, \end{cases} \tag{1.211}$$

where $L > 0$ is the common loss assigned to any incorrect classification. If $\mathbf{x}$ is assigned to class C_j, the corresponding conditional expected loss is

$$\sum_k L_{kj} p(C_k|\mathbf{x}) = L \sum_{k \neq j} p(C_k|\mathbf{x}) = L \left[1 - p(C_j|\mathbf{x})\right], \tag{1.212}$$

Since L is a positive constant and is the same for every candidate class, minimizing the conditional expected loss is equivalent to maximizing $p(C_j|\mathbf{x})$. Therefore, under equal misclassification losses, the decision rule in Eq. 1.210 reduces to Eq. 1.204.

The Reject Option

Classification errors are most likely to occur for inputs $\mathbf{x}$ for which no class has a dominant posterior probability. In such regions, the largest value among $\{p(C_1 \mid \mathbf{x}), \ldots, p(C_K \mid \mathbf{x})\}$ is substantially below one, indicating considerable uncertainty about the true class. Instead of forcing the classifier to make a decision for every input, we can introduce a *reject option*, under which sufficiently uncertain inputs are left unclassified.

Let

$$p_{\max}(\mathbf{x}) = \max_k p(C_k \mid \mathbf{x}) \tag{1.213}$$

denote the largest posterior probability at input $\mathbf{x}$. Given a rejection threshold θ, the classifier makes a class prediction only when $p_{\max}(\mathbf{x}) > \theta$, and rejects the input otherwise. Equivalently, the decision rule can be written as

$$\hat{C}(\mathbf{x}) = \begin{cases} \arg\max_{C_k} p(C_k \mid \mathbf{x}), & \text{if } \max_k p(C_k \mid \mathbf{x}) \geq \theta, \\ \text{reject}, & \text{otherwise.} \end{cases} \tag{1.214}$$

The threshold θ controls the trade-off between the fraction of inputs that can be classified and the uncertainty associated with those decisions. Increasing θ requires greater posterior confidence before a prediction is accepted, so more inputs are rejected but the remaining accepted classifications tend to have a lower error rate. Conversely, decreasing θ results in fewer rejections but allows decisions to be made in more uncertain regions of the input space.

Because the posterior probabilities satisfy

$$\sum_{k=1}^{K} p(C_k \mid \mathbf{x}) = 1, \tag{1.215}$$

the largest posterior probability must satisfy

$$\max_k p(C_k \mid \mathbf{x}) \geq \frac{1}{K}. \tag{1.216}$$

Therefore, choosing $\theta < 1/K$ guarantees that no input is rejected, whereas choosing $\theta = 1$ causes all inputs to be rejected except the degenerate case in which one posterior probability is exactly one.

The reject option can also be incorporated into the minimum-expected-loss framework by treating rejection as an additional action. Let L_{kr} denote the loss incurred when the true class is C_k and the classifier rejects the input. The conditional expected loss of rejection is then

$$R(\text{reject} \mid \mathbf{x}) = \sum_k L_{kr} p(C_k \mid \mathbf{x}). \tag{1.217}$$

For each input $\mathbf{x}$, the classifier compares this quantity with the conditional expected loss of assigning $\mathbf{x}$ to each class,

$$R(C_j \mid \mathbf{x}) = \sum_k L_{kj} p(C_k \mid \mathbf{x}). \tag{1.218}$$

The input is rejected whenever the expected loss of rejection is smaller than that of every class assignment; otherwise, it is assigned to the class with the smallest conditional expected loss. In this way, rejection provides a principled mechanism for avoiding highly uncertain decisions when the cost of abstaining is smaller than the expected cost of making a potentially incorrect classification.

Inference and Decision

We have broken the classification problem down into two separate stages, the *inference* stage, in which training data are used to estimate the posterior probabilities $p(C_k \mid \mathbf{x})$, and a *decision* stage, in which these probabilities are used to make class assignments. Alternatively, the two stages can be combined by learning a function that maps $\mathbf{x}$ directly to a decision. Such a function is called a *discriminant function*.

There are three main approaches to classification, differing in which probabilistic quantities are modeled. These are given in decreasing order of complexity by:

1. The first approach is to model the class-conditional density $p(\mathbf{x} \mid C_k)$ for each class together with the prior class probability $p(C_k)$. The prior $p(C_k)$ can typically be estimated from the relative frequency of class C_k in the training data, Bayes' theorem is then used to obtain the posterior probability,

$$p(C_k \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid C_k) p(C_k)}{\sum_k p(\mathbf{x} \mid C_k) p(C_k)}, \tag{1.219}$$

   Equivalently, we may model the joint distribution $p(\mathbf{x}, C_k)$ directly for each class and normalize it over all classes to obtain $p(C_k \mid \mathbf{x})$. Once the posterior probabilities are available, decision theory is used to determine the class assignment. Approaches that explicitly or implicitly model the joint distribution $p(\mathbf{x}, C_k)$ are called *generative models*. From such a model, synthetic inputs can be generated by first sampling a class from $p(C_k)$ and then sampling from $p(\mathbf{x} \mid C_k)$. Examples of this approach include linear and quadratic discriminant analysis (LDA and QDA), Naive Bayes classifiers, Gaussian mixture classifiers, Bayesian network classifiers, and hidden Markov model classifiers.

2. The second approach is to model the posterior class probabilities $p(C_k \mid \mathbf{x})$ directly, without separately modeling $p(\mathbf{x} \mid C_k)$ and $p(C_k)$. The resulting posterior probabilities are then passed to the decision stage in exactly the same way as before. Models that directly estimate $p(C_k \mid \mathbf{x})$ are called *discriminative models*. Examples of this approach include (multinomial) logistic regression, neural networks with sigmoid or softmax outputs.

3. The third approach removes the probabilistic inference stage altogether. Instead, we learn a *discriminant function* $f(\mathbf{x})$ that maps the input directly to a class label or decision. In this case, the model learns the decision boundary

directly. Examples of this approach include the perceptron, Fisher's linear discriminant, nearest-neighbor classifiers, and nearest-centroid classifiers, which can be trained to separate classes without explicitly modeling class posterior probabilities.

The three approaches differ in both statistical and computational requirements. The first approach is the most demanding because when $\mathbf{x}$ is high-dimensional, accurately estimating $p(\mathbf{x} \mid C_k)$, or equivalently $p(\mathbf{x}, C_k)$ can require a large amount of training data. However, once these distributions are estimated, they also determine the marginal input distribution,

$$p(\mathbf{x}) = \sum_k p(\mathbf{x} \mid C_k) p(C_k). \tag{1.220}$$

This allows the model to identify inputs having very small probability under the training distribution, which can be useful for *outlier detection* or *novelty detection*. However, if the goal is only classification, modeling the full distribution of $\mathbf{x}$ may be unnecessarily difficult because much of the structure in $p(\mathbf{x} \mid C_k)$ may have little effect on $p(C_k \mid \mathbf{x})$. In such cases, directly modeling the posterior probabilities can be more efficient.

The third approach simplifies the problem further by learning a discriminant function that maps $\mathbf{x}$ directly to a class decision, thereby avoiding explicit estimation of both $p(\mathbf{x} \mid C_k)$ and $p(C_k \mid \mathbf{x})$. However, it does not provide posterior probabilities, which are useful for several reasons.

First, posterior probabilities allow the decision rule to be changed without retraining the model. For example, if the loss matrix changes, the new minimum-risk decision can be obtained directly from

$$\hat{C}(\mathbf{x}) = \arg\min_{C_j} \sum_k L_{kj} p(C_k \mid \mathbf{x}). \tag{1.221}$$

whereas a discriminative function may instead need to be retrained.

Second, posterior probabilities support the reject option by quantifying classification uncertainty. Inputs for which the largest posterior probability is too small can be rejected rather than forcibly classified.

Third, posterior probabilities make it possible to adjust for changes in the class priors. Because

$$p(C_k \mid \mathbf{x}) \propto p(\mathbf{x} \mid C_k) p(C_k), \tag{1.222}$$

the training priors can be replaced by the new priors, followed by renormalization across classes. This is useful when the training set has been artificially balanced or when the population class distribution is strongly imbalanced.

Finally, probabilistic outputs make it easier to combine information from multiple models or data sources. Suppose M data sources provide observations $\mathbf{x}_1, \ldots, \mathbf{x}_M$, with one probabilistic model providing the posterior $p(C_k \mid \mathbf{x}_m)$ for each source $\mathbf{x}_m$. Assume that these sources are conditionally independent given the class C_k. The posterior conditioned on all sources is

$$\begin{aligned} p(C_k \mid \mathbf{x}_1, \ldots, \mathbf{x}_M) &\propto p(\mathbf{x}_1, \ldots, \mathbf{x}_M \mid C_k) p(C_k) \\ &= p(C_k) \prod_{m=1}^{M} p(\mathbf{x}_m \mid C_k). \end{aligned} \tag{1.223}$$

Applying Bayes' theorem separately to the output of each model yields

$$p(\mathbf{x}_m \mid C_k) \propto \frac{p(C_k \mid \mathbf{x}_m)}{p(C_k)}. \tag{1.224}$$

Substituting these expressions into the joint posterior gives

$$\begin{aligned} p(C_k \mid \mathbf{x}_1, \ldots, \mathbf{x}_M) &\propto p(C_k) \prod_{m=1}^{M} \frac{p(C_k \mid \mathbf{x}_m)}{p(C_k)} \\ &= p(C_k)^{1-M} \prod_{m=1}^{M} p(C_k \mid \mathbf{x}_m). \end{aligned} \tag{1.225}$$

Thus, if separate probabilistic models provide posterior probabilities from conditionally independent data sources, their outputs can be combined into a joint posterior that incorporates all available sources. This illustrates an important advantage of retaining posterior probabilities rather than learning only a direct input-to-class mapping.

Loss Functions for Regression

Decision theory also applies to regression, where the target t is continuous and the decision is a prediction $y(\mathbf{x})$ for each input $\mathbf{x}$. If predicting $y(\mathbf{x})$ when the true target is t incurs a loss $L(t, y(\mathbf{x}))$, then the expected loss over the joint distribution $p(\mathbf{x}, t)$ is

$$\mathbb{E}[L] = \iint L(t, y(\mathbf{x})) p(\mathbf{x}, t) \, d\mathbf{x} \, dt. \tag{1.226}$$

The goal is to choose the regression function $y(\mathbf{x})$ that minimizes this expected loss. A common choice for $L(t, y(\mathbf{x}))$ is the squared loss,

$$L(t, y(\mathbf{x})) = (y(\mathbf{x}) - t)^2, \tag{1.227}$$

for which

$$\mathbb{E}[L] = \iint (y(\mathbf{x}) - t)^2 p(\mathbf{x}, t) \, d\mathbf{x} \, dt. \tag{1.228}$$

If no restrictions are placed on the form of $y(\mathbf{x})$, its value can be chosen independently for each $\mathbf{x}$. Minimizing the total expected loss reduces to minimizing its contribution separately for each fixed $\mathbf{x}$. For a given $\mathbf{x}$, this contribution is

$$F(y(\mathbf{x})) = \int (y(\mathbf{x}) - t)^2 p(\mathbf{x}, t) \, dt. \tag{1.229}$$

Because $p(\mathbf{x}, t)$ does not depend on $y(\mathbf{x})$, differentiating $F(y(\mathbf{x}))$ with respect to $y(\mathbf{x})$ gives

$$\begin{aligned}\frac{\partial F(y(\mathbf{x}))}{\partial y(\mathbf{x})} &= \int \frac{\partial}{\partial y(\mathbf{x})} \left[(y(\mathbf{x}) - t)^2 \, p(\mathbf{x}, t) \right] \, dt \\ &= \int p(\mathbf{x}, t) \frac{\partial}{\partial y(\mathbf{x})} \, (y(\mathbf{x}) - t)^2 \, dt \\ &= 2 \int (y(\mathbf{x}) - t) \, p(\mathbf{x}, t) \, dt.\end{aligned} \tag{1.230}$$

At the minimum, this derivative is zero. Solving for $y(\mathbf{x})$, and using the sum and product rules of probability, we obtain

$$y(\mathbf{x}) = \int t \, \frac{p(\mathbf{x}, t)}{p(\mathbf{x})} \, dt = \int t \, p(t \mid \mathbf{x}) \, dt = \mathbb{E}[t \mid \mathbf{x}]. \tag{1.231}$$

Thus, under squared loss, the optimal regression function is the conditional mean of the target given the input.

We can also derive this result in a slightly different way, which will also shed light on the nature of the regression problem. The expectation over t with respect to the conditional distribution $p(t \mid \mathbf{x})$ is

$$\begin{aligned}&\mathbb{E}_{t|\mathbf{x}} \left[(y(\mathbf{x}) - t)^2 \right] = \int (y(\mathbf{x}) - t)^2 \, p(t \mid \mathbf{x}) \, dt \\ &= \int \left[(y(\mathbf{x}) - \mathbb{E}[t|\mathbf{x}])^2 + (\mathbb{E}[t|\mathbf{x}] - t)^2 + 2(y(\mathbf{x}) - \mathbb{E}[t|\mathbf{x}])(\mathbb{E}[t|\mathbf{x}] - t) \right] p(t|\mathbf{x}) dt.\end{aligned} \tag{1.232}$$

For a fixed $\mathbf{x}$, both $y(\mathbf{x})$ and $\mathbb{E}[t|\mathbf{x}]$ are constants, and also note that $\int p(t \mid \mathbf{x}) \, dt = 1$. Therefore,

$$\begin{aligned}\mathbb{E}_{t|\mathbf{x}} \left[(y(\mathbf{x}) - t)^2 \right] &= (y(\mathbf{x}) - \mathbb{E}[t \mid \mathbf{x}])^2 + \int (\mathbb{E}[t \mid \mathbf{x}] - t)^2 \, p(t \mid \mathbf{x}) \, dt \\ &\quad + 2 \, (y(\mathbf{x}) - \mathbb{E}[t \mid \mathbf{x}]) \int (\mathbb{E}[t \mid \mathbf{x}] - t) \, p(t \mid \mathbf{x}) \, dt\end{aligned} \tag{1.233}$$

The cross-term vanishes because

$$\begin{aligned}\int (\mathbb{E}[t \mid \mathbf{x}] - t) \, p(t \mid \mathbf{x}) \, dt &= \mathbb{E}[t \mid \mathbf{x}] \int p(t \mid \mathbf{x}) \, dt - \int t \, p(t \mid \mathbf{x}) \, dt \\ &= \mathbb{E}[t \mid \mathbf{x}] - \mathbb{E}[t \mid \mathbf{x}] = 0.\end{aligned} \tag{1.234}$$

The second term is just the conditional variance, i.e., $\mathrm{var}[t \mid \mathbf{x}]$. Hence, for each fixed $\mathbf{x}$,

$$\mathbb{E}_{t|\mathbf{x}} \left[(y(\mathbf{x}) - t)^2 \right] = (y(\mathbf{x}) - \mathbb{E}[t \mid \mathbf{x}])^2 + \mathrm{var}[t \mid \mathbf{x}]. \tag{1.235}$$

Averaging this conditional expected loss over $p(\mathbf{x})$ gives

$$\mathbb{E}[L] = \int (y(\mathbf{x}) - \mathbb{E}[t \mid \mathbf{x}])^2 \, p(\mathbf{x}) \, d\mathbf{x} + \int \mathrm{var}[t \mid \mathbf{x}] p(\mathbf{x}) \, d\mathbf{x}. \tag{1.236}$$

The second integral does not depend on the regression function. It is the conditional variance of t averaged over $\mathbf{x}$ and represents the irreducible uncertainty in the target that can be regarded as noise. Minimizing the expected squared loss reduces to minimizing the first term. Because its integrand is nonnegative for every $\mathbf{x}$, the smallest possible value of the first integral is zero, which is achieved only when $y(\mathbf{x}) = \mathbb{E}[t \mid \mathbf{x}]$ for every $\mathbf{x}$. Hence, we obtain the same optimal regression function as in Eq. 1.231.

As with the classification problem, we can either determine the appropriate probabilities and then use these to make optimal decisions, or we can build models that make decisions directly. Indeed, we can identify three distinct approaches to solving regression problems given, in order of decreasing complexity, by:

1. The first approach is to model the joint density $p(\mathbf{x}, t)$. The conditional density of the target is then obtained by normalization,

$$p(t \mid \mathbf{x}) = \frac{p(\mathbf{x}, t)}{\int p(\mathbf{x}, t) \, dt}. \tag{1.237}$$

   Once $p(t \mid \mathbf{x})$ is available, decision theory is used to determine the regression prediction according to the chosen loss function. For example, under squared loss, the optimal prediction is the conditional mean,

$$y(\mathbf{x}) = \mathbb{E}[t \mid \mathbf{x}] = \int t \, p(t \mid \mathbf{x}) \, dt. \tag{1.238}$$

   As in the classification setting discussed earlier, approaches that explicitly model the joint distribution of the inputs and targets are called generative models. Examples include joint Gaussian models, Gaussian mixture models defined over $(\mathbf{x}, t)$, and joint kernel density estimators.

2. The second approach is to model the conditional density $p(t \mid \mathbf{x})$ directly without modeling the marginal input distribution $p(\mathbf{x})$. Examples of this approach include probabilistic linear regression, heteroscedastic regression models, and mixture density networks, all of which model the distribution of the target conditioned on the input.

3. The third approach removes the probabilistic inference stage altogether. Instead, we learn a regression function $y(\mathbf{x})$ that maps the input directly to a target prediction without explicitly estimating $p(t \mid \mathbf{x})$. Examples include ordinary least-squares regression, neural networks trained with mean squared error, regression trees, and support vector regression.

For the first two approaches, squared loss is not the only possible loss function. A broader family is

$$\mathbb{E}[L_q] = \iint |y(\mathbf{x}) - t|^q \, p(\mathbf{x}, t) \, d\mathbf{x} \, dt. \tag{1.239}$$

For $q = 2$, this reduces to squared loss and the optimal prediction is the conditional mean, i.e., $y(\mathbf{x}) = \mathbb{E}(t \mid \mathbf{x})$ as shown above. For $q = 1$, the optimal prediction is the conditional

median. When $q \to 0$, the optimum approaches the conditional mode. Thus, the optimal regression estimate depends directly on the choice of loss function.

1.11 Information Theory

Information theory provides a quantitative framework for measuring the information obtained from observing random variables. Consider a discrete random variable x with probability mass function $p(x)$. The information associated with observing a particular value of x can be interpreted as its 'degree of surprise'. An unlikely event should provide more information than a likely event; an event that is certain to occur should provide no information.

Let $h(x)$ denote the information associated with observing x. We require $h(x)$ to decrease monotonically as $p(x)$ increases. We also require information to be additive for independent events. If x and y are independent, then $p(x,y) = p(x)p(y)$, while the information obtained by observing both should satisfy $h(x,y) = h(x) + h(y)$. Because the logarithm converts products of probabilities into sums, these requirements lead to the definition of *self-information*

$$h(x) = -\log p(x). \tag{1.240}$$

The negative sign ensures that $h(x) \geq 0$. An event with $p(x) = 1$ has $h(x) = 0$.

The average information obtained by observing the random variable x is the expectation of $h(x)$ with respect to $p(x)$. This quantity is called the *entropy*,

$$\mathrm{H}[x] = \mathbb{E}[h(x)] = -\sum_x p(x) \log p(x). \tag{1.241}$$

For outcomes with $p(x) = 0$, we define their contribution to the entropy to be zero because

$$\lim_{p \to 0} p \log p = 0. \tag{1.242}$$

Entropy can therefore be interpreted as the average amount of information required to specify the outcome of a random variable.

When using logarithms to base 2, the information is measured in *bits*. To understand the motivation of chossing base 2 in information theory by default, consider an example in which x has eight equally likely states. Then $p(x) = 1/8$ for every state, and

$$\mathrm{H}[x] = -8 \left(\frac{1}{8}\right) \log_2 \left(\frac{1}{8}\right) = 3 \text{ bits}. \tag{1.243}$$

This agrees with the fact that eight possible states can be represented exactly using binary strings of length three.

Now consider eight states $\{a, b, c, d, e, f, g, h\}$ with probabilities

$$\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{64}, \frac{1}{64}, \frac{1}{64}, \frac{1}{64}\right). \tag{1.244}$$

The entropy is

$$\begin{aligned} \mathrm{H}[x] = &-\frac{1}{2}\log_2\frac{1}{2} - \frac{1}{4}\log_2\frac{1}{4} - \frac{1}{8}\log_2\frac{1}{8} \\ &- \frac{1}{16}\log_2\frac{1}{16} - 4\left(\frac{1}{64}\log_2\frac{1}{64}\right) \\ &= 2 \text{ bits}. \end{aligned} \tag{1.245}$$

Although there are still eight possible states, the entropy is smaller because the distribution is highly nonuniform. Most of the probability mass is concentrated on a few states, making the outcome more predictable on average. Hence, shorter codewords can be assigned to more probable states and longer codewords to less probable states. The codewords must nevertheless be chosen so that concatenated messages can be decoded unambiguously. For example, the states above may be represented by the codewords

$$0,\ 10,\ 110,\ 1110,\ 111100,\ 111101,\ 111110,\ 111111. \tag{1.246}$$

The expected code length is then

$$\mathbb{E}[\ell] = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{16}(4) + 4\left(\frac{1}{64}(6)\right) = 2 \text{ bits}. \tag{1.247}$$

The relationship between entropy and coding length is formalized by the *noiseless coding theorem*, which states that entropy provides a lower bound on the average number of bits required to encode the outcomes of a random variable using an efficient lossless coding scheme. Hence, entropy is not only a measure of uncertainty, but also a fundamental measure of the amount of information required to represent data.

In machine learning theory, it is often more convenient to use natural logarithms rather than logarithms to base 2. The entropy is then measured in *nats* rather than bits. The two units differ only by the constant conversion factor $\ln 2$.

Using natural logarithms, we can now formalize the pattern observed in the previous two examples: entropy is largest when the probability mass is distributed uniformly across the available states. For a discrete random variable with M possible states,

$$\mathrm{H}[p] = -\sum_{i=1}^{M} p_i \ln p_i, \tag{1.248}$$

subject to the normalization constraint

$$\sum_{i=1}^{M} p_i = 1. \tag{1.249}$$

To find the distribution that maximizes the entropy, we solve a constrained optimization problem with $\mathrm{H}[p]$ as the objective function and the normalization condition on the probabilities as the constraint. Introducing a Lagrange multiplier λ, we maximize

$$\widetilde{\mathrm{H}} = -\sum_{i=1}^{M} p_i \ln p_i + \lambda \left(\sum_{i=1}^{M} p_i - 1 \right). \tag{1.250}$$

Differentiating with respect to the probability vector $\mathbf{p} = (p_1, \ldots, p_M)^T$ gives a gradient vector in $\mathbb{R}^M$, with the ith component given by

$$\frac{\partial \widetilde{\mathrm{H}}}{\partial p_i} = -\ln p_i - 1 + \lambda. \tag{1.251}$$

At a stationary point, the gradient vector must be $\mathbf{0}$, which implies that all probabilities are equal to the same constant $e^{\lambda - 1}$. Applying the normalization constraint gives

$$p_i = \frac{1}{M}, \qquad i = 1, \ldots, M. \tag{1.252}$$

The corresponding maximum entropy is therefore

$$\mathrm{H}_{\max} = -\sum_{i=1}^{M} \frac{1}{M} \ln \frac{1}{M} = \ln M. \tag{1.253}$$

To verify that this stationary point is indeed a maximum, we examine the Hessian of the Lagrangian with respect to the probability vector $\mathbf{p}$. The Hessian is diagonal, with entries

$$\frac{\partial^2 \widetilde{\mathrm{H}}}{\partial p_i \partial p_j} = \begin{cases} -\dfrac{1}{p_i}, & i = j, \\ 0, & i \neq j. \end{cases} \tag{1.254}$$

Because $p_i > 0$ for all i, the Hessian is negative definite. The stationary point is a strict maximum.

We now extend entropy to continuous random variables. Let x be a continuous variable with density $p(x)$. Divide the domain of x into bins of width Δ. If $p(x)$ is continuous, then by the mean value theorem there exists a point x_i in the ith bin such that

$$\int_{i\Delta}^{(i+1)\Delta} p(x)\, dx = p(x_i)\Delta. \tag{1.255}$$

The probability associated with the ith bin is therefore $p(x_i)\Delta$. Treating the bins as the states of a discrete random variable gives

$$\begin{aligned} \mathrm{H}_\Delta &= -\sum_i p(x_i)\Delta \ln \left[p(x_i)\Delta \right] \\ &= -\sum_i p(x_i)\Delta \ln p(x_i) - \ln \Delta, \end{aligned} \tag{1.256}$$

where we have used $\sum_i p(x_i)\Delta = 1$.

As $\Delta \to 0$, the first term approaches a Riemann integral,

$$-\sum_i p(x_i)\Delta \ln p(x_i) \longrightarrow -\int p(x) \ln p(x)\, dx. \tag{1.257}$$

This motivates the definition of the *differential entropy*,

$$\mathrm{H}(x) = -\int p(x) \ln p(x)\, dx. \tag{1.258}$$

The discrete entropy H_Δ differs from the differential entropy by the term $-\ln \Delta$, which diverges as $\Delta \to 0$. Therefore, differential entropy is not the direct limiting value of discrete entropy and, unlike discrete entropy, it can be negative. For a multivariate continuous variable $\mathbf{x}$, the corresponding definition is

$$\mathrm{H}(\mathbf{x}) = -\int p(\mathbf{x}) \ln p(\mathbf{x})\, d\mathbf{x}. \tag{1.259}$$

As in the discrete case, we can ask which continuous distribution has the maximum entropy. Suppose the density is constrained to have mean μ and variance σ^2, the resulting set of three constraints are

$$\int_{-\infty}^{\infty} p(x)\, dx = 1, \tag{1.260}$$

$$\int_{-\infty}^{\infty} x p(x)\, dx = \mu, \tag{1.261}$$

and

$$\int_{-\infty}^{\infty} (x-\mu)^2 p(x)\, dx = \sigma^2. \tag{1.262}$$

We maximize the differential entropy subject to these constraints by introducing three Lagrange multipliers $\{\lambda\}_{i=1}^3$. Because the quantity being optimized depends on the entire density function $p(x)$, the resulting objective is a *functional*, written as

$$\begin{aligned} \widetilde{\mathrm{H}}[p] = &-\int_{-\infty}^{\infty} p(x) \ln p(x)\, dx + \lambda_1 \left[\int_{-\infty}^{\infty} p(x)\, dx - 1 \right] \\ &+ \lambda_2 \left[\int_{-\infty}^{\infty} x p(x)\, dx - \mu \right] + \lambda_3 \left[\int_{-\infty}^{\infty} (x-\mu)^2 p(x)\, dx - \sigma^2 \right]. \end{aligned} \tag{1.263}$$

Collecting all terms that depend on $p(x)$ inside a single integral gives

$$\begin{aligned} \widetilde{\mathrm{H}}[p] = \int_{-\infty}^{\infty} \Big[&-p(x) \ln p(x) + \lambda_1 p(x) + \lambda_2 x p(x) \\ &+ \lambda_3 (x-\mu)^2 p(x) \Big]\, dx - \lambda_1 - \lambda_2 \mu - \lambda_3 \sigma^2. \end{aligned} \tag{1.264}$$

The last three terms do not depend on $p(x)$, so their functional derivatives are zero. Moreover, the integrand depends on $p(x)$ but not on the derivative of $p(x)$. Therefore, the *functional derivative* is obtained by differentiating the integrand directly with respect to $p(x)$ (see Appendix "Calculus of Variations"),

$$\begin{aligned} \frac{\delta \widetilde{\mathrm{H}}}{\delta p(x)} &= -\frac{\partial}{\partial p(x)} \left[p(x) \ln p(x) \right] + \frac{\partial}{\partial p(x)} \left[\lambda_1 p(x) \right] \\ &\quad + \frac{\partial}{\partial p(x)} \left[\lambda_2 x p(x) \right] + \frac{\partial}{\partial p(x)} \left[\lambda_3 (x-\mu)^2 p(x) \right] \\ &= -\ln p(x) - 1 + \lambda_1 + \lambda_2 x + \lambda_3 (x-\mu)^2. \end{aligned} \tag{1.265}$$

The functional must be stationary at the optimial distribution, meaning that its functional derivative is 0 for every x, so

$$p(x) = \exp\left[-1 + \lambda_1 + \lambda_2 x + \lambda_3 (x-\mu)^2\right]. \tag{1.266}$$

Substituting this form back into the three constraints determines the Lagrange multipliers

$$\lambda_1 = 1 - \frac{1}{2}\ln(2\pi\sigma^2), \qquad \lambda_2 = 0, \qquad \lambda_3 = -\frac{1}{2\sigma^2}. \tag{1.267}$$

The density now has the closed-form

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] = \mathcal{N}(x;\mu,\sigma^2). \tag{1.268}$$

Thus, among all continuous distributions with a fixed mean and variance, the Gaussian distribution has the maximum differential entropy. Substituting $p(x)$ into Eq. 1.258, its entropy is

$$\mathrm{H}[x] = \frac{1}{2}\left[1 + \ln(2\pi\sigma^2)\right] = \frac{1}{2}\ln(2\pi e\sigma^2). \tag{1.269}$$

The entropy therefore increases with σ^2, implying that greater uncertainty associated with a broader distribution. This expression also shows that differential entropy can be negative when the variance is sufficiently small,

$$\mathrm{H}[x] < 0 \iff \sigma^2 < \frac{1}{2\pi e}. \tag{1.270}$$

We can also define entropy when part of the information is already known. Suppose $\mathbf{x}$ and $\mathbf{y}$ have a joint distribution $p(\mathbf{x},\mathbf{y})$. Before observing $\mathbf{x}$, the uncertainty in $\mathbf{y}$ is described by its marginal density $p(\mathbf{y})$. Once a particular value of $\mathbf{x}$ is known, however, our uncertainty about $\mathbf{y}$ is described instead by the conditional density $p(\mathbf{y} \mid \mathbf{x})$.

For a fixed value of $\mathbf{x}$, the entropy of this conditional distribution is

$$\mathrm{H}[\mathbf{y} \mid \mathbf{x} = \mathbf{x}_0] = -\int p(\mathbf{y} \mid \mathbf{x}_0) \ln p(\mathbf{y} \mid \mathbf{x}_0)\, d\mathbf{y}. \tag{1.271}$$

Different values of $\mathbf{x}$ generally produce different conditional distributions and therefore different amounts of uncertainty in $\mathbf{y}$. To obtain a single measure of the remaining uncertainty in $\mathbf{y}$ after $\mathbf{x}$ is observed, we average these conditional entropies over all possible values of $\mathbf{x}$ according to $p(\mathbf{x})$,

$$\begin{aligned}\mathrm{H}[\mathbf{y} \mid \mathbf{x}] &= -\int p(\mathbf{x}) \left[\int p(\mathbf{y} \mid \mathbf{x}) \ln p(\mathbf{y} \mid \mathbf{x})\, d\mathbf{y}\right] d\mathbf{x} \\ &= -\iint p(\mathbf{x},\mathbf{y}) \ln p(\mathbf{y} \mid \mathbf{x})\, d\mathbf{x}\, d\mathbf{y}.\end{aligned} \tag{1.272}$$

The conditional entropy $\mathrm{H}[\mathbf{y} \mid \mathbf{x}]$ is the average uncertainty remaining in $\mathbf{y}$ after the value of $\mathbf{x}$ is known.

The joint entropy is

$$\begin{aligned}\mathrm{H}[\mathbf{x},\mathbf{y}] &= -\iint p(\mathbf{x},\mathbf{y}) \ln p(\mathbf{x},\mathbf{y})\, d\mathbf{x}\, d\mathbf{y} \\ &= -\iint p(\mathbf{x},\mathbf{y}) \ln p(\mathbf{y} \mid \mathbf{x})\, d\mathbf{x}\, d\mathbf{y} - \iint p(\mathbf{x},\mathbf{y}) \ln p(\mathbf{x})\, d\mathbf{x}\, d\mathbf{y}\end{aligned} \tag{1.273}$$

The first term is exactly $\mathrm{H}[\mathbf{y} \mid \mathbf{x}]$. For the second term, $\ln p(\mathbf{x})$ does not depend on $\mathbf{y}$, so

$$\begin{aligned}-\iint p(\mathbf{x},\mathbf{y}) \ln p(\mathbf{x})\, d\mathbf{x}\, d\mathbf{y} &= -\int \ln p(\mathbf{x}) \left[\int p(\mathbf{x},\mathbf{y})\, d\mathbf{y}\right] d\mathbf{x} \\ &= -\int p(\mathbf{x}) \ln p(\mathbf{x})\, d\mathbf{x} \\ &= \mathrm{H}[\mathbf{x}].\end{aligned} \tag{1.274}$$

We therefore obtain the entropy chain rule,

$$\mathrm{H}[\mathbf{x},\mathbf{y}] = \mathrm{H}[\mathbf{x}] + \mathrm{H}[\mathbf{y} \mid \mathbf{x}]. \tag{1.275}$$

Thus, the total uncertainty in the pair $(\mathbf{x},\mathbf{y})$ can be viewed as the uncertainty in $\mathbf{x}$ together with the uncertainty that remains in $\mathbf{y}$ after $\mathbf{x}$ has been observed.

Relative Entropy and Mutual Information

Suppose the true distribution of a random variable $\mathbf{x}$ is $p(\mathbf{x})$, but we use another distribution $q(\mathbf{x})$ to represent it. Under an ideal coding scheme based on $p(\mathbf{x})$, an outcome $\mathbf{x}$ is assigned a code length proportional to $-\ln p(\mathbf{x})$. If the code is instead designed using $q(\mathbf{x})$, the same outcome is assigned a code length proportional to $-\ln q(\mathbf{x})$. The difference in code length for that outcome is therefore

$$-\ln q(\mathbf{x}) + \ln p(\mathbf{x}) = \ln \frac{p(\mathbf{x})}{q(\mathbf{x})}. \tag{1.276}$$

Because samples are actually generated according to $p(\mathbf{x})$, the average excess code length is obtained by taking the expectation with respect to $p(\mathbf{x})$,

$$\mathrm{KL}(p\|q) = \int p(\mathbf{x}) \ln \frac{p(\mathbf{x})}{q(\mathbf{x})}\, d\mathbf{x}. \tag{1.277}$$

This quantity is called the *Kullback–Leibler divergence*, or *relative entropy*. It measures the average penalty incurred when $q(\mathbf{x})$ is used in place of the true distribution $p(\mathbf{x})$.

Kullback–Leibler divergence can also be written as

$$\mathrm{KL}(p\|q) = \left[-\int p(\mathbf{x}) \ln q(\mathbf{x})\, d\mathbf{x}\right] - \left[-\int p(\mathbf{x}) \ln p(\mathbf{x})\, d\mathbf{x}\right]. \tag{1.278}$$

The first term is the *cross-entropy* between $p(\mathbf{x})$ and $q(\mathbf{x})$, representing the average code length when samples generated from $p(\mathbf{x})$ are encoded using $q(\mathbf{x})$. The second term is the entropy of the true distribution $p(\mathbf{x})$.

The Kullback–Leibler divergence is also asymmetric,

$$\mathrm{KL}(p\|q) \neq \mathrm{KL}(q\|p), \tag{1.279}$$

so it is not a distance metric.

Another important property of the Kullback–Leibler divergence is its nonnegativity. To establish this result, consider a convex function $f(x)$. For any a and b and any $0 \leq \lambda \leq 1$, convexity requires

$$f\left(\lambda a + (1-\lambda)b\right) \leq \lambda f(a) + (1-\lambda) f(b). \tag{1.280}$$

More generally, for nonnegative weights λ_i satisfying $\sum_i \lambda_i = 1$,

$$f\left(\sum_i \lambda_i x_i\right) \leq \sum_i \lambda_i f(x_i). \tag{1.281}$$

Interpreting the weights as probabilities leads directly to *Jensen's inequality*,

$$f\left(\mathbb{E}[x]\right) \leq \mathbb{E}[f(x)]. \tag{1.282}$$

For a continuous random variable, this becomes

$$f\left(\int x p(x)\, dx\right) \leq \int f(x) p(x)\, dx. \tag{1.283}$$

We now apply Jensen's inequality using the strictly convex function $f(x) = -\ln x$. Rewriting the KL divergence as

$$\mathrm{KL}(p\|q) = \int p(\mathbf{x}) \left[-\ln \frac{q(\mathbf{x})}{p(\mathbf{x})}\right] d\mathbf{x}, \tag{1.284}$$

Jensen's inequality gives

$$\mathrm{KL}(p\|q) \geq -\ln\left[\int p(\mathbf{x}) \frac{q(\mathbf{x})}{p(\mathbf{x})}\, d\mathbf{x}\right] = -\ln 1. \tag{1.285}$$

Therefore,

$$\mathrm{KL}(p\|q) \geq 0. \tag{1.286}$$

Equality holds if and only if

$$p(\mathbf{x}) = q(\mathbf{x}) \tag{1.287}$$

almost everywhere. The KL divergence can therefore be used as a measure of discrepancy between two probability distributions, although its asymmetry prevents it from being a distance.

Kullback–Leibler divergence also provides a direct connection between density estimation and maximum likelihood. Suppose data are generated from an unknown distribution $p(\mathbf{x})$, and we approximate this distribution using a parametric model $q(\mathbf{x} \mid \boldsymbol{\theta})$. We seek parameters that minimize

$$\mathrm{KL}\left(p(\mathbf{x}) \,\|\, q(\mathbf{x} \mid \boldsymbol{\theta})\right). \tag{1.288}$$

Expanding the divergence gives

$$\mathrm{KL}(p\|q) = -\int p(\mathbf{x}) \ln q(\mathbf{x} \mid \boldsymbol{\theta})\, d\mathbf{x} + \int p(\mathbf{x}) \ln p(\mathbf{x})\, d\mathbf{x}. \tag{1.289}$$

Suppose we observe N independent samples $\{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$ drawn from the true distribution $p(\mathbf{x})$. The Kullback–Leibler divergence can be rewritten as an expectation with respect to $p(\mathbf{x})$,

$$\begin{aligned} \mathrm{KL}(p\|q) &= \int p(\mathbf{x}) \ln \frac{p(\mathbf{x})}{q(\mathbf{x} \mid \boldsymbol{\theta})}\, d\mathbf{x} \\ &= \mathbb{E}_{p(\mathbf{x})}\left[\ln p(\mathbf{x}) - \ln q(\mathbf{x} \mid \boldsymbol{\theta})\right]. \end{aligned} \tag{1.290}$$

Because the samples $\mathbf{x}_1, \ldots, \mathbf{x}_N$ are drawn independently from $p(\mathbf{x})$, an expectation under $p(\mathbf{x})$ can be approximated by the corresponding empirical average over the observed samples,

$$\mathrm{KL}(p\|q) \simeq \frac{1}{N} \sum_{n=1}^{N} \left[-\ln q(\mathbf{x}_n \mid \boldsymbol{\theta}) + \ln p(\mathbf{x}_n)\right]. \tag{1.291}$$

Because the term $\ln p(\mathbf{x}_n)$ does not depend on $\boldsymbol{\theta}$. The optimization reduces to

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}} \left[-\frac{1}{N} \sum_{n=1}^{N} \ln q(\mathbf{x}_n \mid \boldsymbol{\theta})\right]. \tag{1.292}$$

Equivalently,

$$\boldsymbol{\theta}^* = \arg\max_{\boldsymbol{\theta}} \sum_{n=1}^{N} \ln q(\mathbf{x}_n \mid \boldsymbol{\theta}) = \arg\max_{\boldsymbol{\theta}} \prod_{n=1}^{N} q(\mathbf{x}_n \mid \boldsymbol{\theta}). \tag{1.293}$$

Thus, choosing the model distribution $q(\mathbf{x} \mid \boldsymbol{\theta})$ that minimizes the KL divergence from the unknown data-generating distribution $p(\mathbf{x})$ can be interpreted as maximum likelihood estimation.

KL divergence can also be used to quantify statistical dependence between two random variables $\mathbf{x}$ and $\mathbf{y}$. If they are independent, their joint distribution factorizes as

$$p(\mathbf{x}, \mathbf{y}) = p(\mathbf{x}) p(\mathbf{y}). \tag{1.294}$$

This motivates defining the *mutual information* as the KL divergence between the joint distribution and the product of the marginals,

$$\begin{aligned} \mathrm{I}[\mathbf{x}, \mathbf{y}] &= \mathrm{KL}\left(p(\mathbf{x}, \mathbf{y}) \,\|\, p(\mathbf{x}) p(\mathbf{y})\right) \\ &= \iint p(\mathbf{x}, \mathbf{y}) \ln \frac{p(\mathbf{x}, \mathbf{y})}{p(\mathbf{x}) p(\mathbf{y})}\, d\mathbf{x}\, d\mathbf{y}. \end{aligned} \tag{1.295}$$

We can also write

$$\mathrm{I}[\mathbf{x}, \mathbf{y}] = \iint p(\mathbf{x}, \mathbf{y}) \left[\ln p(\mathbf{x}, \mathbf{y}) - \ln p(\mathbf{x}) - \ln p(\mathbf{y})\right] d\mathbf{x}\, d\mathbf{y}. \tag{1.296}$$

Since Kullback–Leibler divergence is nonnegative,

$$\mathrm{I}[\mathbf{x}, \mathbf{y}] \geq 0, \tag{1.297}$$

with equality if and only if $\mathbf{x}$ and $\mathbf{y}$ are independent. Mutual information therefore quantifies how far the joint distribution is from the distribution that would result under independence.

The logarithmic term can be written as

$$\ln \frac{p(\mathbf{x}, \mathbf{y})}{p(\mathbf{x})p(\mathbf{y})} = \ln \frac{p(\mathbf{x} \mid \mathbf{y})}{p(\mathbf{x})}. \tag{1.298}$$

Substituting this expression into the mutual information in Eq. 1.296 gives

$$\begin{aligned} \mathrm{I}[\mathbf{x}, \mathbf{y}] = &\left[- \iint p(\mathbf{x}, \mathbf{y}) \ln p(\mathbf{x}) \, d\mathbf{x} \, d\mathbf{y} \right] \\ &- \left[- \iint p(\mathbf{x}, \mathbf{y}) \ln p(\mathbf{x} \mid \mathbf{y}) \, d\mathbf{x} \, d\mathbf{y} \right]. \end{aligned} \tag{1.299}$$

The second term is the conditional entropy $\mathrm{H}[x \mid y]$. The first term can be rewritten as

$$\begin{aligned} - \iint p(\mathbf{x}, \mathbf{y}) \ln p(\mathbf{x}) \, d\mathbf{x} \, d\mathbf{y} &= - \int \ln p(\mathbf{x}) \left[\int p(\mathbf{x}, \mathbf{y}) \, d\mathbf{y} \right] d\mathbf{x} \\ &= - \int p(\mathbf{x}) \ln p(\mathbf{x}) \, d\mathbf{x}, \end{aligned} \tag{1.300}$$

which is the entropy $\mathrm{H}[x]$. Therefore,

$$\mathrm{I}[\mathbf{x}, \mathbf{y}] = \mathrm{H}[\mathbf{x}] - \mathrm{H}[\mathbf{x} \mid \mathbf{y}]. \tag{1.301}$$

Using similar argument but starting instead with $p(\mathbf{x}, \mathbf{y}) = p(\mathbf{y} \mid \mathbf{x})p(\mathbf{x})$, we also obtain

$$\mathrm{I}[\mathbf{x}, \mathbf{y}] = \mathrm{H}[\mathbf{y}] - \mathrm{H}[\mathbf{y} \mid \mathbf{x}]. \tag{1.302}$$

Mutual information can therefore be interpreted as the reduction in uncertainty about one variable obtained by observing the other. Combining these expressions with the entropy chain rule in Eq. 1.275 also gives

$$\mathrm{I}[\mathbf{x}, \mathbf{y}] = \mathrm{H}[\mathbf{x}] + \mathrm{H}[\mathbf{y}] - \mathrm{H}[\mathbf{x}, \mathbf{y}]. \tag{1.303}$$